

Artificial Intelligence for BIM-FM: A theoretical Framework and Future Research Directions

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Abstract

Emerging technologies like Building Information Modelling (BIM), digital twins (DT), and artificial intelligence (AI) are transforming the AEC industry, particularly in the Operation and Maintenance (O&M) phase. This paper explores the potential benefit of integrating these technologies for facility management (FM) stakeholders. A systematic literature review was undertaken to identify state-of-the-art AI techniques, methods, and algorithms used for occupancy modelling, followed by semi-structured interviews with the University of Birmingham's Estates team to better understand their current BIM-FM approach. Key findings identified supervised learning as the predominant approach within the literature with machine learning algorithms such as SVMs, HMM, kNN commonly used for occupancy prediction, and deep learning techniques for vision-based occupancy detection and counting. Interviews highlighted inefficiencies in current occupancy metrics at UoB, based on key identified factors, namely value, effort, and accuracy. Subsequently, a theoretical AI-driven BIM-FM framework was developed, integrating BIM, DT technology, and AI-driven occupancy models to optimise energy and space management at UoB. This theoretical framework was finally proposed to the Estates team through structured interviews, emphasising on its benefits for building management and potential in facilitating decision-making processes, ultimately leading to energy management optimisation, and achieving sustainability goals.

Keywords:

Building Information Modelling (BIM), Artificial Intelligence (AI), Digital Twin (DT), Operation and maintenance (O&M), Facility Management (FM), Occupancy Monitoring

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List of abbreviations

3D: Three-dimensional
4D: Four-dimensional
5D: Five-dimensional
6D: Six-dimensional
7D: Seven-dimensional
AEC - Architecture-engineering construction
AI: Artificial Intelligence
ANN: Artificial Neural Network
BILSTM: Bidirectional Long Short-Term Memory
BIM - Building Information Modelling
BIM-FM: Building Information Modelling - Facility Management
CAD: Computer-aided design
CART: Classification and Regression Trees
CNN: Convolutional Neural Network
COBie: Construction Operations Building Information Exchange
CO2: Carbon dioxide
CV: Computer Vision
DCNN: Deep-Convolutional Neural Network
DT: Digital Twin
DT: Decision Tree
EDMF: Error-Domain Model Falsification
ELM: Extreme Learning Machine
FM - Facility management
GNB: Gaussian Naïve Bayes
GRU: Gated Recurrent Unit
HMM: Hidden Markov Model
IFC: Industry Foundation Classes
ISO: International Organisation for Standardisation
kNN: k-Nearest Neighbour
LGR: Logistic Regression
LSTM: Long Short-term Memory
ML: Machine learning
MPC: Model Predictive Control
NB: Naïve Bayes
NN: Neural Network

O&M - Operation and Maintenance

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

R-CNN: Region-based Convolutional Neural Network

RNN: Recurrent Neural Network

SLR: Systematic literature review

SVM: Support Vector Machine

TAN: Tree Augmented Naïve Bayes

VOC: Volatile Organic Compounds

WSN: Wireless Sensor Network

1. Introduction

With the continuous emergence of new technologies, the construction industry is undergoing significant changes. Historically, the transition from hand-drawn blueprint to computer-aided design (CAD) fundamentally revolutionised the visualisation and planning processes of structures. Over the past decade, Building Information Modelling (BIM) has been successfully adopted in the early stages of a building's lifecycle facilitating organised building information storage, sharing, and exchange among stakeholders, ensuring enhanced collaboration in the architecture, engineering, and construction (AEC) industry (Li et al., 2022). BIM is defined as "a set of interacting policies, processes, and technologies that generate a methodology to manage essential building design and project data in digital format throughout the building's lifecycle" (Succar, 2009). The integration of BIM has resulted in favourable project outcomes across the building's lifecycle (Bryde et al., 2013).

Through its implementation, BIM enables more accurate cost estimation, minimises material waste, and enhances cost control, leading to financial savings and improved management. Additionally, it optimises project scheduling, reduces design errors, and accelerates projects delivery. BIM also allows for better communication and coordination among stakeholders, facilitating improved information sharing, decision-making, and project management efficiency (Firdaus Razali et al., 2019). However, despite the clear benefits that BIM offers in the design and construction phases, its adoption in the later stages, namely the Operation and Maintenance phase (O&M), has been slow and relatively unexplored (Kiviniemi and Codinhoto, 2014).

Up until recently, more scholarly attention has been dedicated to the early phases of the lifecycle due to the immediately tangible outcomes offered by BIM (Pinti et al., 2022). Key stakeholders tend to focus on the initial investments as they occur within a relatively shorter timeframe (typically 2-5 years) as compared to the O&M phase (20 years and beyond) (Abideen et al., 2022). Nevertheless, it is crucial to recognise the significant disparity in cost between these stages. Various sources have indicated that the total cost of the O&M phase far surpasses that of the design and construction phases, with estimations ranging from 60% to 85% of the total lifecycle's cost (Firdaus Razali et al., 2019).

Therefore, as the main contributor of the lifecycle cost the O&M phase has recently demonstrated significant potential for BIM integration, particularly in the field of Facility Management (FM), leading to a growing interest in academic research. FM is defined by the International Organisation for Standardisation (ISO, 2024) 41011:2024 as "an organisational function integrating people, place, and process within the built environment to improve quality of life and business productivity". The integration of BIM in FM streamlines operations by

standardising and centralising building data within a collaborative digital platform which facilitates FM practices, including maintenance, energy management, and other relevant applications (Becerik-Gerber et al., 2012).

However, key issues and limitations hinder its full potential for the O&M phase. A study by Dixit et al. (2019) identified key limitations in BIM integration in FM, notably improper information capture and failure to update BIM data. BIM itself can only provide static information (Tang et al., 2019), which does not adequately reflect the dynamic changes occurring in the built environment over time. These key limitations become notably evident during the later stages of the building lifecycle, where real-time tracking of the building's health is crucial for its management and maintenance (Deng et al., 2021).

To tackle these limitations, various technologies have been explored to integrate real-time data into BIM. The Internet of Things (IoT) characterised by the interconnection of sensing devices capable of exchanging information across different platforms (Gubbi et al., 2013), has shown significant interest over the past years. Through IoT sensing devices, BIM models can incorporate enhanced visualisation and real-time environmental data, facilitating automated updates based on the current status of buildings to ensure the most up-to-date representation of the built environment (Tang et al., 2019). Thus, this integration enables the incorporation of dynamic data alongside the static information provided by BIM models.

The basis of the integration of real-time data into BIM through IoT sensors has led to the emergence of the digital twin (Tang et al., 2019). Digital twin technology is defined as creating a visual and digital model of a physical object (Kaur et al., 2020).

With the emergence of these technologies, it is worth exploring new technologies that can enhance real-time data integration and decision-making processes in the field of FM, notably towards energy optimisation. Buildings are responsible for up to 40% of the global total energy and 30% of greenhouse gas emissions (Cao et al., 2016). As a result, organisations such as the University of Birmingham (UoB) (University of Birmingham, no date) are working towards reducing energy consumption and carbon emissions with the end goal of achieving net zero carbon. This initiative involves the integration of technologies such as BIM, DT, and Building management systems (BMS) towards efficient energy and space utilisation.

2. Literature review

The following section presents a literature review aimed at understanding key aspects discussed such as Building information modelling, digital twins, and artificial intelligence in the AEC industry.

2.1. Building information Modelling

In its quest to decrease project cost, reduce project duration, increase productivity, and improve quality, the AEC industry has explored various technologies, among which BIM stands out as a prominent option (Azhar, 2011). The construction industry is full of nuances, especially on a global level. It is therefore key to explore the various definitions of BIM and understand its fundamental principle. Gu and London (2010) defined BIM as “an IT-enabled approach that involves applying and maintaining an integral digital representation of all building information for different phases of the project lifecycle in the form of a data repository”. This definition emphasised the digitalisation and streamlining of processes within the AEC industry (Borkowski, 2023). However, researchers later emphasised that BIM was much more than just a software or technology, explaining that it is a “virtual process that encompasses all aspects, disciplines, and systems of a facility within a single, virtual model, allowing all team members (owners, architects, engineers, contractors, subcontractors, and suppliers) to collaborate more accurately and efficiently than traditional processes (Azhar et al., 2012).

In this study, a more generalised definition of BIM is adopted, defining BIM as a holistic approach encompassing technology, processes, and collaborative methodologies, aimed at creating and managing digital representations of a built asset throughout its lifecycle. This definition reflects the evolving understanding of BIM’s role and significance within the AEC industry, capturing both its technological aspects and its broader implications for project management and collaboration.

2.1.1. BIM maturity levels and dimensions

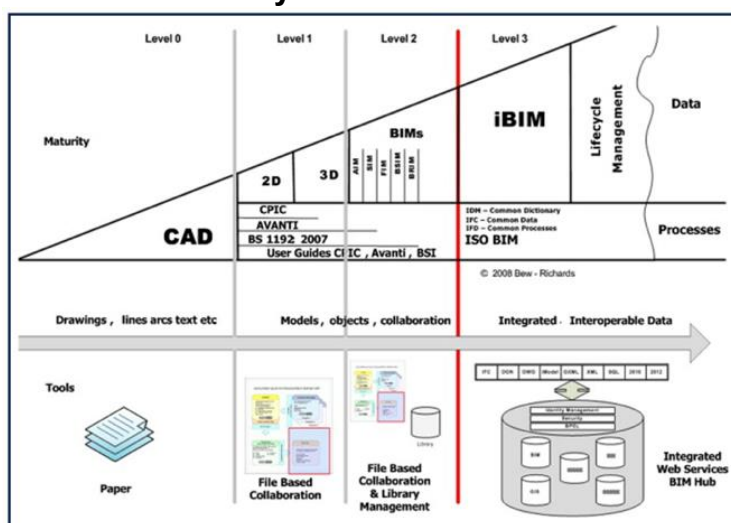


Figure 1. BIM maturity levels (Barlish and Sullivan, 2012)

BIM maturity levels are a way to categorise a project’s proficiency in using BIM technology and processes and are considered as indicators of a construction supply chain’s ability to operate and exchange information (Succar, no date). They range from 0-3 and define what criteria are required to be deemed BIM compliant.

Figure 1 above illustrates the maturity levels of BIM and the associated supported technologies, formats, and tools used. A description of each BIM level is presented below:

“BIM Level 0”: BIM is not employed throughout the building’s lifecycle, relying on traditional methods for information management without digital representation or centralised data exchange between stakeholders (Eischet, 2023).

“BIM Level 1”: Use of BIM tools to create 2D CAD drawings and 3D models for visualisation of building data. Design data is distributed electronically through a Common Data Environment (CDE) (BibLus, 2019), as specified by BS 1192:2007 (Komosko et al., 2017).

“BIM Level 2”: Involves full collaboration between stakeholders and data management across the building’s lifecycle. BIM models are used for 3D visualisation, scheduling, and planning (4D), and cost management (5D) (Eischet, 2023). As of 2016, the UK government mandated BIM level 2 compliance, requiring the adoption of BIM processes and standards throughout construction projects (Abdulkadir and Godfaurd A., 2015). This included creating and managing digital 3D models, collaborative work in a CDE, and using standardised file formats such as the Industry Foundation Classes (IFC), and Construction Operations Building Information Exchange standards (COBie), further detailed in Section 2.1.4.

“BIM Level 3”: Achieves full integration of information within cloud-based environments, enabling the development of a shared model accessible to all stakeholders. This environment allows for real-time interaction and updates, facilitating information of integration throughout the building’s lifecycle. Building upon the foundations established by level 2 BIM, level 3 represents a significant progression towards broader collaboration across the entire lifecycle of the building. The Digital Built Britain strategy (HM Government, 2012), a UK initiative aimed as transforming the construction industry through the integration of digital technologies and processes further outlines the BIM level 3 approach, each seeking to improve BIM capability through new technologies towards an “Open BIM”, an approach aiming at enhancing BIM’s accessibility, usability, management, and sustainability of digital data (buildingSMART International, 2020). The objectives include enabling improvements in the level 2 model, new technologies and systems, and the development of new business models.



Conversely, BIM dimensions represent different levels of information or data integration within a BIM model, with each dimension increasing the model’s complexity. Increasing the dimensions in BIM involves incorporating additional layers of information into the model.

Figure 2. Dimensions of BIM (Josseaux, 2018)

The most common use of BIM is in 3D, which enhances visualisation capabilities. Adding a 4th dimension (4D) introduces the element of time enabling scheduling. Incorporating a 5th dimension (5D) involves including cost-related data, such as material quantities and project budgets. Extending to a 6th dimension (6D) integrates FM information, including maintenance schedules and asset management details. Lastly, including a 7th dimension (7D) encompasses sustainability factors such as energy efficiency and environmental impact (Ellis, 2024).

2.1.2. BIM standards

In May 2011, the UK government released the “BIM Working Party Strategy”, aiming to reduce public sector asset costs through widespread adoption of BIM technology across both public and private sectors involved in buildings and infrastructures delivery. This initiative seeks to enhance operational efficiency, and improve construction supply chain effectiveness (HM Government, 2015). IFC standards (Joseph Paul, 2022) developed by buildingSMART (buildingSMART International, 2019), is an international standard for software interoperability which provides a standardised framework for describing physical objects, geometric representations, and properties within the built environments. The ISO 19650 series (ISO, 2018), defines principles and requirements for information management using BIM throughout an asset’s lifecycle, facilitating the organisation, control, and exchange information across project phases to improve efficiency, reduce errors, and enhance communication among stakeholders. Finally, COBie standards, focusing on non-proprietary data formats, capture “minimum common data” required for maintenance activities for FM in the O&M phase (Kumar and Teo, 2021).

2.2. From BIM to Digital Twins

As previously mentioned, the transition from Building information models to DT has been facilitated by the integration of real-time data into existing 3D models. This integration allows for a transition from static representation to a dynamic virtual representation of a physical asset, allowing for real-time monitoring, analysis, and simulation of their performance (see Section 1). While both require a virtual model, DT emphasises on the existence of a physical object. In addition, a BIM model can represent an asset that hasn’t been built yet, while a DT requires the existence of the physical counterpart (Jiang et al., 2021). By incorporating live data streams from sensors, IoT devices, and other sources into the DT environment, stakeholders can gain valuable insights into the current state of assets, predict future behaviour, and optimise decision-making processes. DTs are increasingly being used in the AEC industry (Zhang et al., 2023), and have shown increasing scholarly attention in various fields such as smart cities, bridges, and wastewater, roads, and railway among others (Naderi and Shojaei, 2022; Jiménez Rios et al., 2023). To meet the growing demand in DTs (Zhang et al., 2023), especially in the O&M phase, various platforms have been developed.

Among them, Autodesk Tandem stands out as a cloud-based digital twin platform which enables collaboration across the industry. It leverages BIM data and presents it back to stakeholders, within the context of an accurate virtual model (Autodesk Tandem, no date).

It should be noted that UoB Estates is a recognised leader in BIM and DT adoption in the UK through its initiative to develop a smart campus by digital twinning, using Autodesk Tandem (Hopper, et al., 2024).

2.3. Artificial intelligence

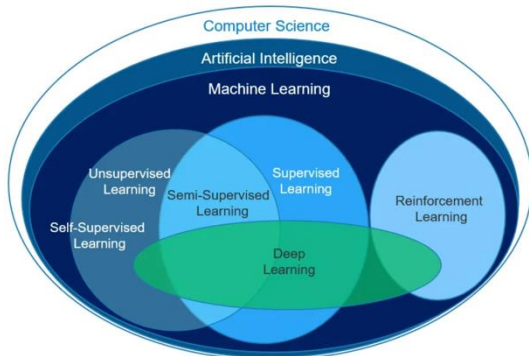


Figure 3. AI and its subsets (Mysla, 2020)

Artificial Intelligence (AI) encompasses the science and engineering of creating intelligent machines that exhibit reasoning, learning, knowledge, communication, perception, planning, and the ability to move and operate objects (McCarthy, 1956). AI offers several benefits, including reducing human error through predictive analysis, efficiently handling Big Data, and facilitating

decision-making (Kumar, 2019; Maheshwari, 2023). Due to these advantages, AI has played a crucial role in industrialising processes across various industries, including healthcare (Jiang et al., 2017; Teo et al., 2024) and manufacturing (Yu et al., 2024; Yamamoto et al., 2024), among others. More recently, the AEC industry has also recognised AI's potential (Darko et al., 2020) due to the need for more efficient information processing. The volume and complexity of data generated across a building's lifecycle now exceeds human's processing capabilities, making AI an essential prospect for optimising processes in the AEC sector. In FM notably, AI can streamline operations through automation and enable data-driven decision-making, optimising resource allocation and energy consumption (Prasad, 2023). The subsets of AI notably include machine learning (ML), Computer Vision (CV), Experts systems, Natural Processing languages and various others. This study will specifically focus on ML and CV for occupancy modelling.

ML aims to enable computers or systems to "learn" and "improve" from experience without directly being explicitly programmed and involves developing algorithms and models capable of analysing data, identify patterns, and making predictions based on that data (Mahesh, 2018; Bi et al., 2019). As noted by El Boucheffy and de Souza (2020), ML involves teaching algorithms using a training dataset, consisting of independent variables (input) and corresponding dependent variables (output). The algorithm learns the relationship between the input and output data during training. Once trained, the algorithm can make predictions on new input data by applying this learned relationship. Testing the algorithm with a separate dataset assesses its predictive accuracy. It can be subdivided into various subsets (as illustrated in Figure 3) based on the types of learning, including supervised learning,

unsupervised learning, and semi-supervised learning (Wakefield, 2019). The three types of ML are described below.

Supervised learning: Supervised learning relies on labelled datasets to train algorithms to classify data and predict outcomes accurately (Wang, Cui and Ke, 2023). It involves providing the model with input-output pairs, allowing it to measure its accuracy and improve over time.

Unsupervised learning: Unsupervised learning uses unlabelled, unclassified, and categorised training data with the main goal of discovering hidden and relevant patterns (El Bouchefry and de Souza, 2020). It has various applications for cluster analysis, with object recognition as the most relevant for this study.

Semi-supervised learning: Semi-supervised learning combines aspects of both supervised and unsupervised learning (El Bouchefry and de Souza, 2020). They use partially labelled training data, typically including a small set of labelled data along with a larger pool of unlabelled data. The algorithms are trained using both sets of data, which enhances their learning accuracy significantly.

BIM, DT and AI have the potential to further impact the AEC industry, and it is evident that their integration will tremendously benefit stakeholders involved in FM. This research will therefore explore various ML algorithms that can potentially be implemented within a new theoretical BIM-FM framework at UoB. A theoretical framework is “a foundational review of existing theories that serves as a roadmap for developing the arguments you will use in your own work” (Vinz, 2022).

3. Aim and Objectives

The aim of this study is to develop a new theoretical AI-driven BIM-FM framework, to enable optimisation of facilities performances through occupancy monitoring and space management at UoB, using identified state-of-the-art AI techniques, methodologies, and algorithms. Through this aim, this study will provide critical insights to enable the Estates team to achieve UoB’s goals towards reducing energy consumption and carbon emissions with the end goal of achieving net zero carbon.

The associated objectives are to:

1. Undertake a critical systematic literature review to identify key AI techniques, methods, and algorithms, and key components needed to develop an AI-driven occupancy model.
2. Employ unstructured interviews with the UoB Estates team to gain key insights on their current BIM-FM best practice approach, identifying and evaluating current occupancy monitoring metrics based on factors such as value, effort, and accuracy.
3. Develop a novel AI-driven BIM-FM framework tailored to the requirements of the Estates team, integrating existing technologies with the developed AI-driven occupancy model.

4. Employ semi-structured interviews to propose and validate the new AI-driven BIM-FM framework, underscoring its potential in achieving UoB's sustainability goals through its implementation.

4. Methodology

The study employed three key methodologies: a systematic literature review, semi-structure interviews, and structured interviews.

4.1. Systematic Literature Review

A Systematic Literature Review (SLR) approach was adopted to identify AI techniques relevant to the aim of this study, following the PRISMA-2020 guidelines which provided guidance on the protocol for conducting the SLR (Page et al., 2021). An SLR is a structured research method used to collect, identify, and critically analyse existing literature (e.g., articles, conference papers, reviews) on a specific topic through a systematic, unbiased, and transparent manner (Tranfield, D. et al., 2003). The SLR was organised into five stages: formulating the research questions, identifying relevant work, assessing the quality of studies, summarising the evidence, and interpreting the findings.

4.1.1. Research Questions:

To ensure a thorough examination of the AI techniques for occupancy monitoring, the following research questions were formulated to guide the systematic literature review:

- 1: What AI techniques, methods, or models have been used for occupancy monitoring?
- 2: How do they compare based on key criteria such as effort, value, and accuracy?
- 3: How can they be implemented to enhance the current BIM-FM approach at UoB and optimise energy consumption?

4.1.2. Database Selected

The primary literature database identified for this SLR was Scopus by Elsevier. Scopus is the largest abstract and citation scientific databases (Harzing and Alakangas, 2016), offering the highest reliability in comparison with other databases (e.g. Web of Science). It provides a comprehensive coverage across multiple disciplines and ensuring access to a wide range of relevant scholarly literature. It offers an extensive collection of peer-reviewed journals, reviews, conference proceedings, and other academic sources which facilitates a thorough exploration of the research topic. Moreover, Scopus offers advanced search filtering capabilities, allowing a precise retrieval of relevant literature based on specific criteria set for the search, such as the source type, publication stage or source type.

4.1.3. Literature Identification

Using the Scopus search tool, a focused exploration of AI-driven occupancy models was undertaken, focusing on AI-related terms such as "Machine Learning", "Deep Learning", "Neural Network", and "Computer Vision". The search aimed to identify document relevant to

real-time data integration and AI-driven decision-making processes for occupancy monitoring in FM. The final Scopus search string through Boolean operation is shown in Table 5 (see Section 9, Appendix 3).

The initial unfiltered search returned 121 documents, which were subsequently refined by narrowing down the publications between 2013 and 2024 to focus the search on the most recent studies from the past decade, limiting document types to articles, conference papers, and reviews, and excluding non-English documents and literature in preliminary stages of publication. The filtering process resulted in a final dataset of 105 documents prior to the screening process, which were organised and classified using the Mendeley reference management software (Mendeley, 2024) to facilitate further analysis during the screening process and referencing.

Following the initial filtering stage, a screening process was undertaken to eliminate documents irrelevant to the research aim. This involved a comprehensive review of all titles and abstracts and a full-text analysis (presented in Section 9, Appendix 3). Additionally, during the full-text assessment phase both Forward and Backward Snowballing techniques (Thomé et al., 2016) were used to ensure a thorough examination of the literature. Forward snowballing involved identifying documents cited within the initially identified literature. Conversely, backward snowballing involved searching for relevant literature in the reference list of the initially identified documents, which may not have been identified during the initial database search. The SLR process is illustrated in Section 9 (Appendix 3, Figure 3).

4.2. Interviews

Following the SLR, semi-structured interviews were conducted with the Estates team (see Table 1) to gather essential insights on the UoB's BIM-FM best practice approach. Semi-structured interviews are recognised as one of the most common and effective interviews techniques (Deakin university, 2022). They involve the interviewer preparing a set of questions and topics in advance (see Section 9, Appendix 4), which can be asked in various ways to different interviewees. This method enhances flexibility and responsiveness from participants while maintaining the interview's focus, ensuring reliability and credibility in data collected.

Subsequently, structured interviews with the Estates team (see Section 9, Appendix 4) were conducted to propose, evaluate, and validate the implementation of the new AI-driven BIM-FM framework at UoB. The interview process for both semi-structured and structured interviews followed the guidelines outlined in Cohen, Manion and Morrison (2017) for appropriate planning, conducting, and reporting of structured interviews. Additionally, in accordance with the University of Birmingham's "Code of Practice for Research" (University of Birmingham, no date), the interview process underwent a full ethical review (see Section 9, Appendix 1), and consent was collected from each participant (see Section 9, Appendix 2).

4.2.1. Sampling strategy and participants profiles

A purposive sampling strategy (Omona, 2013) was used to identify and select executive-level participants from the UoB Estates team and gather valuable insights into operational processes and requirements related to the proposed new AI-driven-FM framework. The experience in relevant fields of participants (see Table 1) enabled a comprehensive understanding of the team's key requirements and existing challenges.

Table 1. Estates team participants' profiles

ID	Role/job description	Years of experience
R.D.	Head of BIM and Digital Assets	20 years in Building Management
M.H.	Data manager/BIM and Digital Assets	8 years in Digital information Management
C.P.	Space Manager/BIM and Digital Assets	17 years in Space Management

5. Results

5.1. Systematic literature review

The results of this section involved analysing 35 documents through full-text analysis. Several techniques and algorithms were identified and presented in Table 3. In addition, several data sources were identified, such as ambient sensors, Wi-Fi networks, and video cameras (see Table 2 below).

Table 2. Data sources identified in the SLR.

Data source	Data collected	Mentioned in
Wireless ambient Sensors	Environmental data such as temperature, CO2 concentration, light, Volatile Organic Compounds (VOC), humidity, acoustics, air pressure, Indoor Air Quality (IAQ),	Ortega et al., 2015; Yang et al., 2014; Wang et al., 2018; Varnosfaderani et al., 2021; Stewart et al., 2016; Dong and Lam, 2011; Chen et al., 2016; Saralegui et al., 2017; Irvan et al., 2018; Wang et al., 2017; Wang et al., 2021; Alishahi et al., 2021;
3D depth sensors CCTV cameras	Depth data, camera frames, captured images, labelled images, video frames	Diraco et al., 2015; Antonino et al., 2019; Tien et al., 2022; Tien et al., 2020; Tien et al., 2021; Dino et al., 2022; Wei et al., 2022; Hu et al., 2023;
Wi-Fi network	Wi-fi signal	Wang et al., 2018; Tien et al., 2020; Wang et al., 2017; Chen et al., 2019; Alishahi et al., 2021;
Smart meters	Advanced metering infrastructure	Feng et al., 2020;

The main data sources identified for occupancy monitoring include wireless ambient sensors, 3D depth sensors, CCTV cameras, Wi-Fi networks, and smart meters. These

sources provide a range of environmental data such as temperature CO2 concentration, light, and humidity, as well as images and video frames.

Table 3. Identified AI techniques and algorithms used of occupancy modelling

AI subset		Algorithm identified
Machine Learning	Supervised learning	Support Vector Machine (SVM)
		K-nearest neighbours (kNN)
		Hidden Markov Model (HMM)
		Naïve Bayes (NB)
		Decision Tree (DT)/CART
		Tree Augmented Naïve Bayes Network (TAN)
		Neural Network
		Artificial Neural Network (ANN)
		Convolution Neural Network (CNN)
		(Faster) Region-based CNN (R-CNN)
		Recurrent Neural Network (RNN) (LSTM /BiLSTM)
		Gated Recurrent Unit (GRU)
		Extreme Learning Machine (ELM)
		Gaussian Naïve Bayes (GNB)
		Logistic Regression (LGR)
		Random Forest
		Domain Knowledge
		k-mean clustering
		PDT Bayesian model
		Extreme Gradient Boosting
Computer Vision/Sensor Fusion	Supervised Learning	YOLO (You Only Look Once)
		Region-based Convolutional Neural Network
	Sensors	3D depth sensors SR4000
		3D depth sensors Kinect sensors
		IMR sensors
		Wireless Sensor Network (WSN)

Refer to Section 9, Appendix 5 for full table including references.

5.1.1. Identified algorithms for occupancy monitoring

Occupancy monitoring can be subdivided into two primary categories: occupancy detection and occupancy prediction. Occupancy detection focuses on monitoring occupancy at a current state, while occupancy prediction involves forecasting occupancy levels at a future state (Ryu and Moon, 2016). The SLR identified supervised learning as the most used ML approach for occupancy monitoring, with the main algorithms used for occupancy prediction included SVMs, kNN, HMM, Neural Networks, and Random Forest (as shown in Table 3 above). Conversely, deep learning techniques and algorithms such as Convolutional Neural Networks (CNN), using 3D depth sensors and CCTV cameras to collect images and video frames.

Support Vector Machine (SVM): SVMs are described as ML algorithms used for binary classification tasks, such as detecting occupancy status in buildings (Varnosfaderani et al., 2021). SVMs have been used in various studies to predict occupancy by using ambient sensors data such as temperature, humidity, light intensity, and motion and have achieved high accuracy (Yang et al., 2014; Varnosfaderani et al., 2021). In Ortega et al. (2015) notably, an experimental evaluation was conducted, indicating that SVM outperformed other methods such as kNN and HMM in terms of accuracy and consistency, with an average accuracy of 71.64% across different scenarios.

Hidden-Markov Models: HMMs are statistical algorithms that represent a system with hidden states and observable outputs. They are used to predict occupancy based on observed data, assuming that the system's state is not directly visible but can be inferred from observable outputs. A study by Ryu and Moon (2016) conducted experiments using an HMM model based on indoor and outdoor CO₂ concentrations and lighting systems electricity consumption. The results showed that the environmental data-driven model using HMM algorithms performed well in predicting and detecting occupancy.

Vision-based approach:

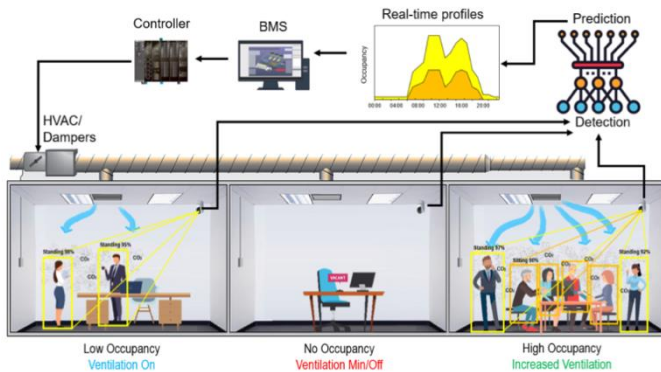


Figure 4. Vision-based approach framework (Wei et al., 2022)

In various studies (Feng et al., 2020; Tien et al., 2022; Dino et al., 2022; Mahajan et al., 2023), computer vision-based occupancy models using deep learning algorithms were developed and tested to detect occupancy. In Wei et al. (2022) notably, a vision-based approach framework was developed for demand-based ventilation control

(see Figure 4 above). The framework used AI-enabled cameras to collect real-time occupancy data, generates an occupancy profile, and inputs it into the Building Management System (BMS) to automatically adjust HVAC system operations, thus ensuring efficient building energy performance and maintaining user's comfort. A Faster RCNN model was trained to detect the number of people and occupancy activity. RCNN work by combining region proposals with convolutional neural networks and classify objects within an image. The process involves extracting regions of interest from the input data, creating region-based features from a pre-trained CNN, and then classifying the objects through SVMs for example.

5.1.2. Key findings

Key results in the SLR demonstrated the accuracy of proposed AI-driven occupancy models in detecting and predicting occupancy. In addition, studies highlighted the significant potential of using these models in reducing energy consumption.

In Wang et al. (2018), on-site experiments were conducted to validate the accuracy of a developed occupancy model, and the results demonstrated the effectiveness of the model in accurately capturing occupancy distribution. In Hu et al. (2023), a novel occupancy model used DCNN for feature extracting from CCTV and a three-stage detection procedure to calculate the number of occupants and identify their locations. The approach demonstrated superior performance in occupancy detection, particularly in complex scenarios with multiple occupants. The study compared the proposed approach with various other methods using various metrics such as F1 score, Precision, Mean Square Error, and others.

Many studies have also highlighted the significance of AI-driven occupancy monitoring for reducing energy consumptions. Indeed, Ortega et al. (2015) emphasised on the impact of user behaviour on energy consumption in buildings. Wang et al. (2018) also demonstrated the potential benefits of AI-driven approaches in building energy management. Long-term insights on occupancy can help facility managers identifying energy peak demand periods of buildings, allowing stakeholders to make informed decisions to reduce energy consumption. Tien et al. (2021) conducted an experiment to further demonstrate the potential of AI-driven occupancy model in energy savings. The experiment involved a vision-based occupancy model and achieved 97.32% accuracy in detecting occupants' activities. Simulations then indicated that the process allowed significant reduction heating and cooling energy consumption.

Various studies have assessed the use of different data sources in occupancy modelling accuracy. Environmental sensor data have notably been used for building occupancy estimation by capturing changes in occupancy through the inherent relationship between occupancy and environmental parameters such as CO₂, temperature, humidity, and air pressure (Chen et al., 2016). In Yang et al. (2014), CO₂, door status, and light were identified as the most influential sensor variables, while average sound, net door count, and infrared count showed less success. Varnosfaderani et al. (2021), evaluated different environmental parameters such as CO₂, volatile organic compounds (VOC), light, temperature, and humidity through various statistical models based on SVMs, Random Forest, KNN and Logistic Regression. The results indicated that VOC can sometimes work better than background CO₂ for detecting occupancy levels, noting however that the performance of VOC as an indicator was greatly influenced by the characteristics and activities of occupants, and the presence of food and other product which may affect the VOC levels.

In addition, a study by Arief-Ang et al., (2018) indicated that although CO2 was identified as reliable metric for binary detection of occupancy (e.g. vacant or occupied room) with 93.71% accuracy, it showed low accuracy in predicting the exact number of occupants.

Hu et al. (2023), used CCTV cameras, offering richer occupancy data than other sensors. The approach involved using Deep Convolutional Neural Network (DCNN) for feature extraction and a three-stage occupancy detection process, achieving high performance in detection occupants and their locations, with empirical experiments validating its effectiveness. However, it should be noted that privacy issues arise when using camera-based systems, particularly regarding the collection of private data. To address privacy-related issues, another study (Antonino et al., 2019) developed a system by placing the cameras vertically on the ceiling towards the floor to avoid capturing faces. In addition, the system is unable to recognise people, and images are not stored but rather used for real-time detection of occupancy.

5.2. Interviews' results

5.2.1. Crucial factors identified (R.D.)

Informal interviews (summarised in appendix C) with R.D., explored three key factors when selecting AI techniques for occupancy monitoring: “value”, “effort”, and “accuracy”. These factors are crucial for determining the most suitable ML techniques for integration into the proposed new BIM-FM framework. Discussions initially recognised the importance of practicality and cost through effort required in implementing and maintaining AI systems. Additionally, the value proposition of the identified ML approach was examined, focusing on their ability to address specific challenges and deliver tangible benefits. The interviews finally addressed the optimal level of accuracy required to make significant impact on outcomes and informed decision-making.

5.2.2. Data management (M.H.)

In discussions with M.H., valuable insights were gained into current BIM-F approach, including the effectiveness of the current occupancy metrics. Challenges related with adopting Autodesk Tandem were identified, including the lack of coding expertise within the estate's team and insufficient funding from the University administration to explore new technologies. The introduction of Tandem Connect, an integration platform allowing the connection of live data streams into Tandem's digital twin, was also discussed. Additionally, challenges related to the lack of resources availability and budget constraints were highlighted as potential obstacles to implementing new technologies within the current framework. It was recommended to conduct initial testing of the proposed framework on a smaller scale to gather empirical data and assess its value, before considering full-scale implementation. This empirical data will serve as motivation for the administration to allocate resources to the proposed framework.

5.2.3. Space management (C.P.)

Following an interview with C.P., a better understanding of the current system used by the estates team to assess occupancy was developed. Firstly, the importance of space type when considering the three key factors discussed in section 5.2.1. Buildings on UoB campus are classified into three main categories: research-based, teaching-based, and support-based. Next, the discussion focused on the how the utilisation of a building is calculated. C.P. explained that the calculation of a building utilisation is determined by the product of frequency (how often a building is being used) and occupancy (the level of occupancy during use). Finally, the primary data streams currently used for occupancy monitoring within the existing approach were explored. These include Wi-Fi data, lighting systems (in collaboration with Siemens) and attendance monitoring through the MyUoB application.

6. Discussion

The following section firstly discusses the current approach used by Estates at UoB for occupancy monitoring, including the main stakeholders involved and an evaluation of the current metrics used for occupancy monitoring. The new AI-driven approach is then developed, with an overview of the theoretical framework, its main components, and the AI-driven occupancy model.

6.1. Current Approach BIM-FM

6.1.1. Stakeholders

Three main stakeholders have been identified, namely the Estates management team, the students at the University, and the University administration. Estates managers oversee the physical infrastructure of the university, ensuring operational efficiency and compliance with regulations. University students as primary users of campus facilities prioritise user experience, learning environments, convenience, and accessibility. Additionally, the administration allocates resources effectively. Balancing the interests of these stakeholders is crucial for an effective building management, optimising facilities to meet student comfort while maintaining operational effectiveness and financial sustainability.

6.1.2. Current metrics used for occupancy monitoring

The metrics identified by R.D, M.H., and C.P. for assessing building occupancy included lighting sensors, WI-FI data, and attendance through MyUoB app. Each metric offers insights into building occupancy but also presents its own set of challenges and limitations.

These metrics were evaluated based on the three key factors identified by R.D., namely the value proposition, accuracy, and effort for implementation (see Section 5.2.1). They were assessed on a scale from 1-5 to determine their effectiveness and efficiency.

Value: This metric assesses the usefulness of feedback provided by the occupancy data. A rating of 1 indicates inefficient feedback that fails to offer timely and valuable insights into occupancy levels. Conversely, a rating of 5 means valuable feedback that provides detailed information on space utilisation, allowing stakeholders to make effective decisions based on real-time data, such as reallocating lecture theatres or temporarily closing buildings.

Effort: Reflects the cost and disruption associated with implementing and maintaining the metric system. A rating of 1 means extensive costs and significant disruption to daily student and staff activities during implementation. Conversely, rating of 5 suggests low costs and minimal disruption, ensuring smooth integration into existing buildings without major interruptions to student and staff activities.

Accuracy: This metric assesses the system's precision in determining occupancy. A rating of 1 signifies basic activity detection without specifying when or where it occurred. Conversely, a rating of 5 indicates high accuracy, enabling precise occupancy counts in designated areas at specific times over defined periods.

Table 4. Current metrics evaluation according to value, effort, and accuracy, rated by R.D.

Metric	Value	Effort	Accuracy
Lighting sensors	1 (Short term) – 3 (long term)	5	3-4
Wi-Fi	4	2	3
MyUoB app Attendance	3-4	1	2

The value proposition ranged from 1 to 3, with a rating of 1 indicating low value in the short term as feedback on occupancy is provided two days after data collection, while there is moderate value in the long term, as highlighted by R.D. (see Section 9, Appendix 4), with the system's ability to give valuable insights on space utilisation in the long term. Effort scored 1 due to extensive costs involved in implementation and significant disruptions to students and staff activities, as the lighting system was completely replaced. Finally, lighting sensors only inform on activity detection over a period without providing an occupancy count.

Wi-Fi has high potential in terms of value as the system is being revisited to operate at a room level rather than building. In terms of accuracy, the metric system give insights on how many people are connected to the server but does not indicate their location.

Finally, attendance monitoring through MyUoB app requires minimal effort as schools across the university already have it in place and is not intrusive in any way. However, the feedback provided may not always be reliable, as students can manipulate results by sharing attendance codes. Consequently, this metric may not always accurately reflect the actual occupancy count within the buildings.

6.2. Theoretical AI-driven BIM-FM framework proposal

6.2.1. Overview

The proposed AI-driven BIM-FM framework aims to enhance and extend the current datasets of the Estates team by integrating additional metrics that offer deeper insights into university space utilisation. This will involve implementing more accurate occupancy detection techniques to monitor present occupancy levels, and predictive models to forecast future occupancy trends based on environmental data, using the identified AI algorithms (see Section 5.1.2). In addition, the framework will capitalise on existing technologies already in use in the current approach, such as BIM and digital twins to establish a solid collaborative platform.

This platform will be used as a space where various disciplines within the Estates team, including data, space, and sustainability management can collaborate more effectively, ensuring better communication and coordination among stakeholders. In addition, it will enable the administration, responsible for critical decision-making related to space utilisation, energy optimisation and sustainability, to better visualise key metrics such as energy consumption more effectively.

6.2.2. Key Components

Building Information modelling: BIM serves as a key component in the framework, both as a technology and process (as explained in Section 2.1.1), enabling efficient collaboration and information exchange among stakeholders. BIM will also serve as a centralised data repository for FM-related static data.

Autodesk Tandem: The digital twin platform is used to integrate real-time data from sensors to monitor key ambient data such as air quality, CO₂ levels, volatile Organic Compounds (VOC), lighting, temperature, and humidity. Data streams will be integrated through Tandem Connect discussed in Section 2.2.2.

IoT sensor: IoT sensors will be deployed across campus to gather real-time data. This process begins with selecting appropriate sensors based on the specific data type needed, including light, CO₂, VOC, and temperature sensors, as discussed in Section 6.2.2. Once deployed, these sensors will continuously collect data from their respective environments. Collected data will then be processed by the AI-driven occupancy model and extract relevant insights on occupancy.

6.2.3. Data collection and integration

Based on the results presented in Section 5.1.2, the new framework will include existing and new metrics as data sources. VOC have shown to be reliable indicators of occupancy despite VOC levels affected by factors such as the presence of food and other occupant's activities. However, it should be noted that in key research-based rooms, food and smoking is prohibited. Other environmental data that will be integrated include Lighting (already in use), temperature,

and CO2 levels. These metrics have shown to be reliable indicators for occupancy. CCTV cameras will also be used due to their high detection accuracy and rich occupancy data.

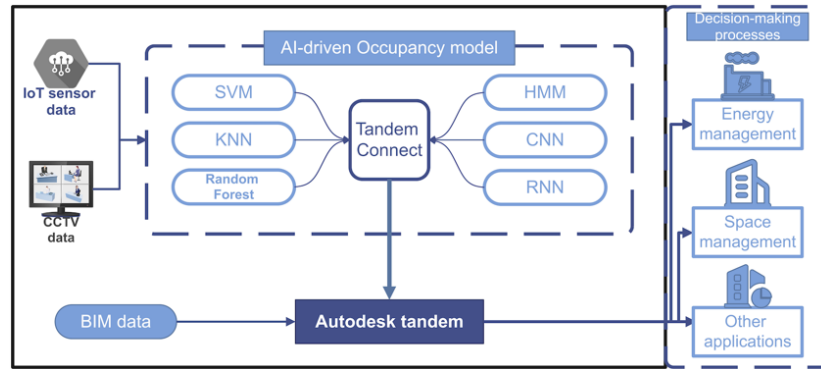


Figure 5. Functionality-based theoretical AI-driven BIM-FM framework.

Figure 5 illustrates how the functionality-based framework workflow. Live data from sensors and cameras are collected from the built environment and processed by the AI-driven occupancy model to detect occupancy at current and future state. Occupancy output is then transferred to the DT platform through tandem Connect. In addition, key environmental data are also directly integrated and presented into the DT to enable better visualisation of these key parameters. Information stored in the DT platform is then shared, exchanged, and used to enable decision-making processes towards energy and space management.

AI-driven occupancy prediction model:

Occupancy prediction will be achieved by an occupancy model based on four identified ML algorithms, namely SVM, kNN, HMM, and Neural Networks. As discussed in Section 5.1.1, these algorithms showed the highest accuracy and performance in occupancy monitoring. Combining them into an occupancy model has many benefits, including complementing their strengths, and improving accuracy through ensemble methods. The model also provides flexibility to adapt to various scenarios such as varying data distribution.

SVMs showed high performance in binary classification tasks due to their ability to find the best possible boundary between data, making them particularly useful for complex datasets. HMMs, can capture pattern changes over time such as predicting occupancy levels throughout the day. kNN is straightforward and offers simplicity and efficiency with non-parametric characteristics. Finally, Neural Networks, particularly CNNs, are good at learning complex patterns from various sources such as images and sensor data.

AI-driven Occupancy detection model:

The algorithm selected to process CCTV data is primarily based on the novel deep-learning-based approach for occupancy detection and localisation, proposed by Hu et al. (2023). Experimental results of the proposed model showed superior performance on occupancy detection and localisation when compared with other methods. The approach used a deep

CNN for feature extraction and implemented a three-stage occupancy detection procedure (refer to section 5.1.1).

6.2.4. Challenges and future research directions

Through discussions with the Estates team, a key challenge in the implementation of the theoretical framework was identified. M.H. and R.D. indicated that the Estates team currently lacks funding and has a limited budget dedicated to adopting new technologies. In addition, the Estates team currently does not have trained personnel who are familiar with data analytics and machine learning programming. However, it is important to highlight the potential of this framework to reduce energy consumption, which would have a positive impact in terms of cost. Indeed, several studies have shown great cost benefits with implementing AI-driven occupancy monitoring models. In addition, as previously mentioned, UoB has sustainability goals to reach, and the implementation of this framework has great potential to be a key factor in achieving this goal. Finally, as a research-based institution, UoB has the potential to establish itself as a leader in AI-driven technologies and set a benchmark for best practices among UK universities. To further support the framework's proposal, future research should prioritise developing, training, and testing the AI-driven occupancy models. Additionally, conducting pilot studies including experiments on the UoB campus will provide empirical data to further support findings from existing literature. This will then allow for the development of a business case to secure additional funding. In addition, research should also focus on developing a model capable of automating control systems based on occupancy output, as discussed in Section 5.1.1.

6.3. Framework's validation

The AI-driven BIM-FM framework, including the AI-driven occupancy model, was presented to the Estate's team as an alternative approach to their BIM-FM strategy (Appendix 4). R.D. and M.H. showed enthusiasms for the implementation of AI technology, citing current trends towards the adoption of new technologies. As compared to the current approach used for occupancy, the proposed theoretical framework shows better accuracy in occupancy detection and prediction enabling feedback on building's occupancy status and facilitating actions for space utilisation and energy optimisation. Through the adoption of this framework, the Estates team will have a better insight on buildings occupancy and utilisation on campus, thus enabling them to make informed decisions on space and resource allocations. In the long term, this will enable reductions in energy consumptions, and potentially greenhouse emissions and carbon footprints towards achieving UoB's sustainability goals. R.D. and M.H. approved the framework (Section 9, Appendix 4) and are considering adopting AI technologies discussed in this report. R.D indicated that this is the direction that needs to be adopted to achieve total digitalisation of campus, into a smart campus.

7. Conclusion

In conclusion, the integration of artificial intelligence into BIM-FM shows great potential for enhancing building management practices and decision-making processes at the University of Birmingham. The initial objectives set at the start of the research have all been achieved.

A systematic literature review was conducted to identify suitable AI techniques and algorithms to develop an AI-driven occupancy model aimed at detecting and predicting occupancy levels across the UoB campus. The review revealed that machine learning was the most used technique such as Support Vector Machines, Hidden Markov Models, K-Nearest Neighbours, and Convolutional Neural Networks performed the best in predicting occupancy based on environmental data. Environmental parameters such as lighting, temperature, VOC, and CO₂ were selected as primary occupancy metrics based on their effectiveness. In addition, deep learning algorithms such as Convolutional Neural Networks were identified as the most suitable approach to detect occupancy, using data collected from depth sensors and CCTV cameras. Then, unstructured interviews with UoB Estates were undertaken, allowing a better understanding of their current BIM-FM best practice approach. Key metrics used for occupancy monitoring were identified, namely Wi-Fi, lighting sensors, and student attendance, and evaluated based on key identified factors, namely the value proposition, accuracy, and effort involved in implementation. Key findings identified limitations and inefficiencies in the current metrics used for occupancy monitoring.

Following the first set of interviews, a novel BIM-FM framework was developed, tailored to the requirements of the Estates team. The framework integrates existing technologies already in use at UoB, such as Building Information Modelling and digital twin technologies, with BIM used for collaborative processes and digital twins as a data repository and platform for live data. It also integrates new components such as IoT sensors for real-time, dynamic data integration, and a new AI-driven occupancy model to process occupancy-related data collected and provide feedback on building occupancy status. This will facilitate decision-making processes related to space utilisation and energy optimisation.

Finally, semi-structured interviews with M.H. and R.D. from the Estates team to propose the new AI-driven BIM-FM framework, supported by research-based evidence highlighting its potential benefit to the current approach. The implementation of an AI-driven occupancy model could enable reductions in energy consumptions and can be used for various energy management applications such as HVAC control.

The theoretical framework's potential was finally validated by the Estates team who will consider recommendations and future research directions such as designing the AI-driven occupancy model and test it on campus to assess its effectiveness based on empirical data.

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9. Appendices

Appendix 1: Full Ethical Review

UNIVERSITY OF BIRMINGHAM Department of Civil Engineering APPLICATION FOR ETHICAL REVIEW

Who should use this form:

This form is to be completed by students who have completed the Research Project Outline and have decided that further ethical review and approval is required before the commencement of a given Research Project.

This form is based on the University Research Ethics Review form and should be used for BEng, MEng and MSc projects (only) which have identified that a full ethical review is required.

[PGR projects should seek central university ethics approval using the webpages below]

NOTES:

- Answers to questions must be entered in the space provided.
- An electronic version of the completed form should be submitted to the module coordinator by email. Please **do not** submit paper copies.
- If, in any section, you find that you have insufficient space, or you wish to supply additional material not specifically requested by the form, please submit it in a separate file, clearly marked and attached to the submission email.
- The elements of the project covered by the ethical review (e.g. questionnaire or interviews) **cannot** proceed until written approval of the ethical review has been received.
- If you have any queries about the form, please address them to the module coordinator.



Before submitting, please tick this box to confirm that you have consulted and understood the following information and guidance and that you have taken it into account when completing your application:

- The information and guidance provided on the University's ethics webpages (<https://intranet.birmingham.ac.uk/finance/accounting/Research-Support-Group/Research-Ethics/Ethical-Review-of-Research.aspx>)
- The University's Code of Practice for Research (<http://www.birmingham.ac.uk/Documents/university/legal/research.pdf>)

UNIVERSITY OF BIRMINGHAM
Department of Civil Engineering
APPLICATION FOR ETHICAL REVIEW

1. TITLE OF PROJECT

Artificial Intelligence for BIM-FM: A Novel Framework and Future Research Directions

2. INVESTIGATORS

a) PLEASE GIVE DETAILS OF THE SUPERVISOR(S)

Name:	Richard Davies
School/Department	School of Engineering, Department of Civil Engineering
Email address:	r.j.davies.1@bham.ac.uk

b) PLEASE GIVE DETAILS OF THE STUDENT

Name of	Khalil Ibrahim	Student No:	2035444
Course of study:	Meng Civil Engineering	Email	KXI945@student.bham.ac.uk

3. ESTIMATED START OF PROJECT

Date: 27/09/23

ESTIMATED END OF PROJECT

Date: 21/03/24

4. SUMMARY OF PROJECT

Describe the purpose, background rationale for the proposed project, as well as the hypotheses/research questions to be examined and expected outcomes. This description should be in everyday language that is free from jargon. Please explain any technical terms or discipline-specific phrases.

The objective of this study is to investigate the potential integration of artificial intelligence (AI) techniques within the operational and maintenance phases of building management, particularly within the framework of Building Information Modelling (BIM). Through a systematic literature review, I aim to identify relevant AI technologies that can be used for real-time data integration and/or AI-driven decision-making and propose a novel framework for their application in BIM-FM contexts. Subsequently, I plan to engage professionals in the field through interviews to assess the feasibility, limitations, and potential of my proposed approach.

5. CONDUCT OF PROJECT

Please give a description of the research methodology that will be used

Through a systematic literature review, existing studies and frameworks will be analysed to establish a foundational understanding of current practices and identify current AI techniques being used in the industry. Subsequently, a novel framework for implementing AI techniques within the BIM-FM context will be developed based on the insights collected from the literature review. This framework will serve as a new approach for using AI to optimise operational and maintenance processes in buildings and more specifically, energy consumption at the university of Birmingham. Additionally, interviews with professionals and experts in the fields of BIM, FM, and AI within the University of Birmingham will be conducted to gather valuable insights, perspectives, and feedback on the proposed framework. Transcripts of the recorded interviews will carefully be reviewed to extract relevant insights and perspectives from participants. A software program called NVivo will be used to analyse the qualitative data collected.

6. DOES THE PROJECT INVOLVE PARTICIPATION OF PEOPLE OTHER THAN THE RESEARCHERS AND SUPERVISORS?

Yes ☒ No ☐

Note: "Participation" includes both active participation (such as when participants take part in an interview or return a questionnaire) and cases where participants take part in the study without their knowledge and consent at the time (for example, in crowd behaviour research).

If you have answered NO please go to Section 14. If you have answered YES to this question please complete all the following sections.

7. PARTICIPANTS AS THE SUBJECTS OF THE RESEARCH

Describe the number of participants and important characteristics (such as age, gender, location, affiliation, level of fitness, intellectual ability etc.). Specify any inclusion/exclusion criteria to be used.

The exact number of participants is still unknown. Potential participants for the interviews were selected using a sampling strategy based on their expertise and roles at the university of Birmingham administration or asset management team. Participants outside of the university may also be interviewed depending on their level of expertise.

8. RECRUITMENT

Please state clearly how the participants will be identified, approached and recruited. Include any relationship between the investigator(s) and participant(s) (e.g. instructor-student).

Note: Attach a copy of any poster(s), advertisement(s) or letter(s) to be used for recruitment.

To identify and recruit participants for the study, potential participants meeting the criteria mentioned above will be identified through established networks within the University community and on LinkedIn. These individuals will then be approached through verbal communication or email.

9. CONSENT

a) Describe the process that the investigator(s) will be using to obtain valid consent. If consent is not to be obtained explain why. If the participants are minors (children) or for other reasons are not competent to consent, describe the proposed alternate source of consent, including any permission / information letter to be provided to the person(s) providing the consent.

Consent will be obtained using a participant Information sheet which includes a consent form.

Note: Attach a copy of the Participant Information Sheet, the Consent Form, the content of any interview script (if applicable) and any other material that will be used in the consent process.

b) Will the participants be deceived in any way about the purpose of the study? Yes ☐ No ☒

If yes, please describe the nature and extent of the deception involved. Include how and when the deception will be revealed, and who will administer this feedback.

N/A

10. PARTICIPANT FEEDBACK

Explain what feedback/ information will be provided to the participants after participation in the research. (For example, a more complete description of the purpose of the research, or access to the results of the research).

Participants will be offered to receive a copy of the final report.

11. PARTICIPANT WITHDRAWAL

a) Describe how the participants will be informed of their right to withdraw from the project.

The participants will be informed of their withdrawal options through the Participant Information Sheet and will be verbally reminded at the beginning of the interview.

b) Explain any consequences for the participant of withdrawing from the study and indicate what will be done with the participant's data if they withdraw.

Data collected from participants who wish to withdraw for the research will be deleted. However, participants will be informed that withdrawal from the study after the 18th of March will not be possible through the information sheet as removing their data from the report may take time.

12. COMPENSATION

Will participants receive compensation for participation?

i) Financial (i.e. payment)

Yes ☐ No ☒

ii) Non-financial (i.e. gifts or other inducements)

Yes ☐ No ☒

If **Yes** to **either** i) or ii) above, please provide details.

N/A

If participants choose to withdraw, how will you deal with compensation?

N/A

13. CONFIDENTIALITY

a) Will all participants be anonymous?

Yes ☐ No ☒

b) Will all data be treated as confidential?

Yes ☐ No ☒

Note: Participants' identity/data will be confidential if an assigned ID code or number is used, but it will not be anonymous. Anonymous data cannot be traced back to an individual participant.

Describe the procedures to be used to ensure anonymity of participants and/or confidentiality of data both during the conduct of the research and in the release of its findings.

It is important for this research to identify experts and professionals whose opinions and input is valuable. Therefore, their job titles and years of experience may be added in the report. Their consent will be taken through the Participant Information Sheet. However, their names will not be mentioned in the report, instead they will be identified using letters (e.g. A, B, C, ...)

If participant anonymity or confidentiality is not appropriate to this research project, explain, providing details of how all participants will be advised of the fact that data will not be anonymous or confidential.

Through the Participant Information Sheet.

14. STORAGE, ACCESS AND DISPOSAL OF DATA

Describe what research data will be stored, where, for what period of time, the measures that will be put in place to ensure security of the data, who will have access to the data, and the method and timing of disposal of the data.

Interviews will be recorded to aid note taking. Audio files will be stored in original format with written transcripts being produced to accompany the report. The data will only be stored on the student investigator and supervisor's UoB network (One Drive) that require a password. After graduation, the data collected will be deleted from all devices and accounts.

15. ARE ANY OTHER APPROVALS REQUIRED?

e.g. Criminal Records (DBS) checks, Animal or Medical information approval or Company approval for commercially confidential information

☐

YES

☒

NO

If yes, please specify whose approval is required and how this will be obtained.

N/A

16. RISKS

a) Outline any potential risks to **INDIVIDUALS**, including research staff, research participants, other individuals not involved in the research and the measures that will be taken to minimise any risks and the procedures to be adopted in the event of mishap

N/A

b) Outline any potential risks to **THE ENVIRONMENT and/or SOCIETY** and the measures that will be taken to minimise any risks and the procedures to be adopted in the event of mishap.

N/A

17. ARE THERE ANY OTHER ETHICAL ISSUES RAISED BY THE RESEARCH?

Yes ☒ No ☐

If yes, please specify

Yes, restricted information may be shared by the University administration. If so, approval will be sought for this "restricted data" to be used in the report. One of the participants is the head of Assets and BIM at the University of Birmingham.

A separate form will be sent to the head of assets to ask for his permission.

18. CHECKLIST

Please mark if the study involves any of the following:

- Vulnerable groups, such as children and young people aged under 18 years, those with learning disability, or cognitive impairments ☐
- Research that induces or results in or causes anxiety, stress, pain or physical discomfort, or poses a risk of harm to participants (which is more than is expected from everyday life) ☐
- Risk to the personal safety of the researcher ☐
- Deception or research that is conducted without full and informed consent of the participants at time study is carried out ☐
- Administration of a chemical agent or vaccines or other substances (including vitamins or food substances) to human participants. ☐
- Production and/or use of genetically modified plants or microbes ☐
- Results that may have an adverse impact on the environment or food safety ☐
- Results that may be used to develop chemical or biological weapons ☐

Please check that the following documents are attached to your application.

	ATTACHED	NOT APPLICABLE
Recruitment advertisement	☐	☒
Participant information sheet	☒	☐
Consent form	☒	☐
Questionnaire	☐	☒
Interview Schedule/Questions/Script	☒	☐

19. DECLARATION BY APPLICANTS

I submit this application on the basis that the information it contains is confidential and will be used by the University of Birmingham for the purposes of ethical review and monitoring of the research project described herein, and to satisfy reporting requirements to regulatory bodies. The information will not be used for any other purpose without my prior consent.

I declare that:

- The information in this form together with any accompanying information is complete and correct to the best of my knowledge and belief and I take full responsibility for it.
- I undertake to abide by University Code of Practice for Research (http://www.as.bham.ac.uk/legislation/docs/COP_Research.pdf) alongside any other relevant professional bodies' codes of conduct and/or ethical guidelines.
- I will report any changes affecting the ethical aspects of the project to the module coordinator.
- I will report any adverse or unforeseen events which occur to the module coordinator.

Name of student:

Khalil Ibrahim

Date:

05/02/24

Please now save your completed form, print a copy for your records, and then email a copy to the module co-ordinator for approval. As noted above, please do not submit a paper copy.

Appendix 2: Signed Information Sheets and Consent Forms

MEng Final Year Project

Information and Consent Form

Project Title: Artificial Intelligence for BIM-FM: A Novel Framework and Future Research Directions

Student investigator: Khalil Ibrahim (email: KXI945@student.bham.ac.uk, Tel: 07478119389)

Supervisor: Dr Richard Davies (email: r.j.davies.1@bham.ac.uk)

The University of Birmingham guidelines stipulate that an ethical review must be conducted for any project collecting information that could be deemed sensitive. As part of this procedure, this form details the information that is being collected and how it will be stored and used. Should any further information be required, contact details for the student investigator are provided above.

What information is being collected? Professional insights will be collected regarding the implementation of AI in BIM-FM for real-time data integration and AI-driven decision-making. This research aims to propose a novel framework, and interviews will be conducted to gather professional opinions on its feasibility and potential impact. Participants personal information such as job title and/or position on the Estates team at the University of Birmingham may be included in the report.

How will the information be used? The information gathered through this interview will serve as critical input for assessing the feasibility of the proposed framework and exploring potential future research directions. Insights obtained will provide valuable perspectives on the practicality, challenges, and opportunities associated with implementing AI in BIM-FM for real-time data integration and decision-making. By analysing this input, the study aims to evaluate the viability of the proposed framework in addressing the objectives and goals set by the Estates team for facility management. The minutes of this interview will be recorded and included in the appendix of the report. Information may also be included in the body of the report.

How will the information be stored and deleted? Interviews may be recorded to aid notetaking, with audio files being stored in their original format. Written transcripts will be produced to accompany the report. The data will only be stored on the research author's UoB OneDrive account on a password-protected computer. Data will be retained until graduation after which it will be securely deleted from all devices and storage locations.

Can I retract my data from the project? Yes. Should you no longer wish for your data to be used, please get in touch **before the 18th of March** and any data will be deleted.

I agree to any information I disclose in this interview to be recorded and used in the research project entitled "Artificial Intelligence for BIM-FM: A Novel Framework and Future Research Directions".

Signed: XXXXXXXXXXXXXXXX **Date:** 11/03/24

Name: XXXXXXXXXXXXXXXX

MEng Final Year Project

Information and Consent Form

Project Title: Artificial Intelligence for BIM-FM: A Novel Framework and Future Research Directions

Student investigator: Khalil Ibrahim (email: KXI945@Student.bham.ac.uk, Tel: 07478119389)

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I agree to any information I disclose in this interview to be recorded and used in the research project entitled "Artificial Intelligence for BIM-FM: A Novel Framework and Future Research Directions".

Signed: XXXXXXXXXXXXXXX

Date: 21/03/2024

Name: XXXXXXXXXXXXXXX

MEng Final Year Project

Information and Consent Form

Project Title: Artificial Intelligence for BIM-FM: A Novel Framework and Future Research Directions

Student investigator: Khalil Ibrahim (email: KXI945@Student.bham.ac.uk, Tel: 07478119389)

Supervisor: Dr Richard Davies (email: r.j.davies.1@bham.ac.uk)

The University of Birmingham guidelines stipulate that an ethical review must be conducted for any project collecting information that could be deemed sensitive. As part of this procedure, this form details the information that is being collected and how it will be stored and used. Should any further information be required, contact details for the student investigator are provided above.

What information is being collected? Professional insights will be collected regarding the implementation of AI in BIM-FM for real-time data integration and AI-driven decision-making. This research aims to propose a novel framework, and interviews will be conducted to gather professional opinions on its feasibility and potential impact. Participants personal information such as job title and/or position on the Estates team at the University of Birmingham may be included in the report.

How will the information be used? The information gathered through this interview will serve as critical input for assessing the feasibility of the proposed framework and exploring potential future research directions. Insights obtained will provide valuable perspectives on the practicality, challenges, and opportunities associated with implementing AI in BIM-FM for real-time data integration and decision-making. By analysing this input, the study aims to evaluate the viability of the proposed framework in addressing the objectives and goals set by the Estates team for facility management. The minutes of this interview will be recorded and included in the appendix of the report. Information may also be included in the body of the report.

How will the information be stored and deleted? Interviews may be recorded to aid notetaking, with audio files being stored in their original format. Written transcripts will be produced to accompany the report. The data will only be stored on the research author's UoB OneDrive account on a password-protected computer. Data will be retained until graduation after which it will be securely deleted from all devices and storage locations.

Can I retract my data from the project? Yes. Should you no longer wish for your data to be used, please get in touch **before the 18th of March** and any data will be deleted.

I agree to any information I disclose in this interview to be recorded and used in the research project entitled "Artificial Intelligence for BIM-FM: A Novel Framework and Future Research Directions".

Signed: XXXXXXXXXXXXXXXX

Date: 18/03/24

Name: XXXXXXXXXXXXXXXX

Appendix 3: Systematic Literature review

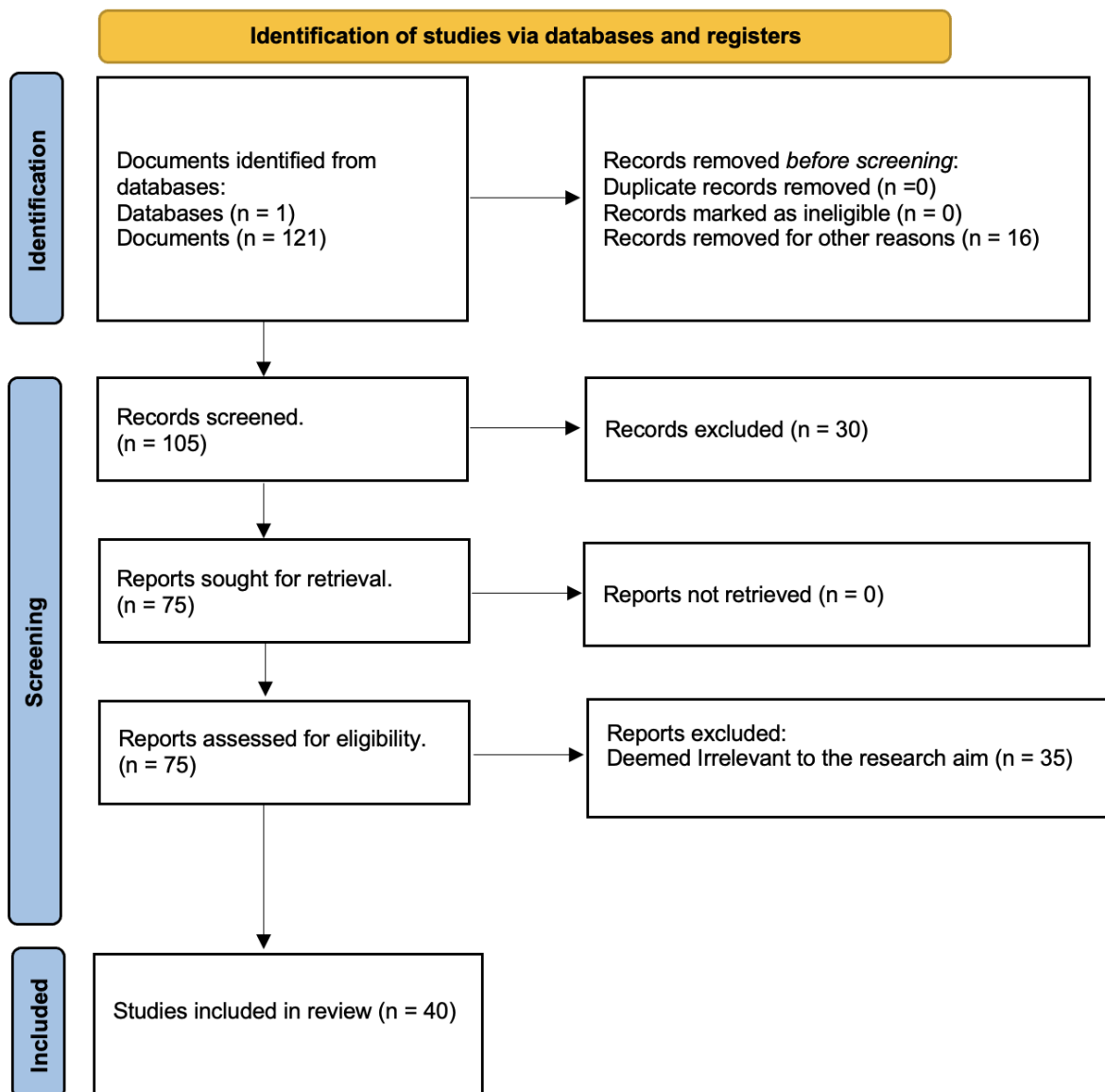


Figure 6. PRISMA 2020 flow diagram for systematic reviews

Table 5. Final Scopus search through Boolean operations

FINAL SEARCH STRING THROUGH BOOLEAN OPERATION
(TITLE-ABS-KEY (building AND information AND modelling) OR TITLE-ABS-KEY (building AND information AND modeling) OR TITLE-ABS-KEY (building AND information AND model) OR TITLE-ABS-KEY (bim) AND TITLE-ABS-KEY (artificial AND intelligence) OR TITLE-ABS-KEY (ai) OR TITLE-ABS-KEY (machine AND learning) OR TITLE-ABS-KEY (supervised AND learning) OR TITLE-ABS-KEY (unsupervised AND learning) OR TITLE-ABS-KEY (naïve AND bayes) OR TITLE-ABS-KEY (case AND based AND reasoning) OR TITLE-ABS-KEY (decision AND tree) OR TITLE-ABS-KEY (k-nearest AND neighbors) OR TITLE-ABS-KEY (logistic AND regression) OR TITLE-ABS-KEY (artificial AND neural AND networks) OR TITLE-ABS-KEY (deep AND learning) OR TITLE-ABS-KEY (convolutional AND neural AND networks) OR TITLE-ABS-KEY (genetic AND algorithms) OR TITLE-ABS-KEY (natural AND processing AND language) OR TITLE-ABS-KEY (reinforcement AND learning) OR TITLE-ABS-KEY (data AND mining) OR TITLE-ABS-KEY (time AND series AND analysis) OR TITLE-ABS-KEY (generative AND adversarial AND networks) OR TITLE-ABS-KEY (predictive AND analytics) OR TITLE-ABS-KEY (pattern AND recognition) OR TITLE-ABS-KEY (fuzzy AND logic) OR TITLE-ABS-KEY (big AND data) OR TITLE-ABS-KEY (internet AND of AND things) AND TITLE-ABS-KEY (data AND integration) OR TITLE-ABS-KEY (real-time AND data) OR TITLE-ABS-KEY (real-time AND data AND integration) OR TITLE-ABS-KEY (dynamic AND data) OR TITLE-ABS-KEY (dynamic AND data AND integration) OR TITLE-ABS-KEY (smart AND meters) OR TITLE-ABS-KEY (iot AND sensors) OR TITLE-ABS-KEY (decision AND making) OR TITLE-ABS-KEY (data AND processing) AND TITLE-ABS-KEY (energy AND optimisation) OR TITLE-ABS-KEY (dynamic AND building AND control) OR TITLE-ABS-KEY (predictive AND energy AND management) OR TITLE-ABS-KEY (energy AND efficiency) OR TITLE-ABS-KEY (space AND management) OR TITLE-ABS-KEY (occupancy AND monitoring) AND TITLE-ABS-KEY (facility AND management) OR TITLE-ABS-KEY (facilities AND management) OR TITLE-ABS-KEY (fm) OR TITLE-ABS-KEY (bim-fm) OR TITLE-ABS-KEY (operation AND maintenance) OR TITLE-ABS-KEY (building AND facility) OR TITLE-ABS-KEY (smart AND building) OR TITLE-ABS-KEY (smart AND facility) OR TITLE-ABS-KEY (smart AND city) OR TITLE-ABS-KEY (smart AND campus) OR TITLE-ABS-KEY (digital AND shadow) OR TITLE-ABS-KEY (digital AND twin)) AND PUBYEAR > 2015 AND PUBYEAR < 2024 AND (LIMIT-TO (SUBJAREA, "COMP") OR LIMIT-TO (SUBJAREA, "ENGI") OR LIMIT-TO (SUBJAREA, "ENER") OR LIMIT-TO (SUBJAREA, "DECI") OR LIMIT-TO (SUBJAREA, "MATH")) AND (LIMIT-TO (SRCTYPE, "j") OR LIMIT-TO (SRCTYPE, "p") OR LIMIT-TO (SRCTYPE, "k")) AND (LIMIT-TO (LANGUAGE, "English")) AND (LIMIT-TO (PUBSTAGE, "final"))

Table 6. Complete SLR results with references.

AI subset identified	Technique identified	References
Machine Learning	Support Vector Machine (SVM)	Ortega et al., 2015; Yang et al., 2014; Varnosfaderani et al., 2021; Chen et al., 2016; Irvan et al., 2018; Feng et al., 2020; Alsmirat et al., 2019; Wang et al., 2021; Esrafilian-Najafabadi and Haghighat, 2021; Gursel Dino et al., 2022; Drira and Smith, 2022; Alishahi et al., 2021; Nair et al., 2023; Roda-Sanchez et al., 2024; Hu et al., 2023; Kim and Moon, 2024
	K-nearest neighbours (kNN)	Gómez Ortega et al., 2015; Yang et al., 2014); (Varnosfaderani et al., 2021; Chen et al., 2016; Feng et al., 2020; Chen et al., 2019; Wang et al., 2021; Esrafilian-Najafabadi and Haghighat, 2021; Kim and Moon, 2024
	Hidden Markov Model (HMM)	Gómez Ortega et al., 2015; Ryu and Moon, 2016; Saralegui et al., 2017; Irvan et al., 2018; Esrafilian-Najafabadi and Haghighat, 2021; Alishahi et al., 2021; Kim and Moon, 2024
	Naïve Bayes (NB)	Yang et al., 2014
	Decision Tree (DT)/CART	Yang et al., 2014; Wang et al., 2021; Gursel Dino et al., 2022; (Roda-Sanchez et al., 2024); Nair et al., 2023; Ryu and Moon, 2016; Chen et al., 2016
	Tree Augmented Naïve Bayes Network (TAN)	Yang et al., 2014
	Neural Network	Saralegui et al., 2017; Irvan et al., 2018; Chen et al., 2019
	Artificial Neural Network (ANN)	Yang et al., 2014; Chen et al., 2016; Dino et al., 2022; Alishahi et al., 2021
	Convolution Neural Network (CNN)	Tien et al., 2022; Tien et al., 2020); Feng et al., 2020; Dino et al., 2022; Wei et al., 2022; Hu et al., 2023; Kim and Moon, 2024
	(Faster) Region-based CNN (R-CNN)	Tien et al., 2022; Tien et al., 2021; Wei et al., 2022; Hu et al., 2023; Kim and Moon, 2024
	Recurrent Neural Network (RNN) (LSTM /BiLSTM)	Feng et al., 2020; Esrafilian-Najafabadi and Haghighat, 2021; Dino et al., 2022; Hu et al., 2023; Kim and Moon, 2024
	Gated Recurrent Unit (GRU)	Esrafilian-Najafabadi and Haghighat, 2021
	Extreme Learning Machine (ELM)	Wang et al., 2018; Chen et al., 2016
	Gaussian Naïve Bayes (GNB)	Varnosfaderani et al., 2021
	Logistic Regression (LGR)	Varnosfaderani et al., 2021; Alsmirat et al., 2019; Roda-Sanchez et al., 2024
	Random Forest	Varnosfaderani et al., 2021; Alsmirat et al., 2019; Wang et al., 2021; Esrafilian-Najafabadi and Haghighat, 2021; Roda-Sanchez et al., 2024; Kim and Moon, 2024
	Domain Knowledge	Wang et al., 2021
	k-mean clustering	Alishahi et al., 2021
	PDT Bayesian model	Stewart et al., 2016
Computer Vision/Sensor Fusion	3D depth sensors SR4000	Diraco et al., 2015
	3D depth sensors Kinect sensors	Diraco et al., 2015
	IMR sensors	Antonino et al., 2019
	Wireless Sensor Network (WSN)	Brooks et al., 2014
Other	Model Predictive Control (MPC)	Brooks et al., 2014
	Probabilistic model	Chen et al., 2019
	Error-Domain Model Falsification (EDMF)	Drira and Smith, 2022
	Stochastic random walk	Chen et al., 2019
	Extreme Gradient Boosting	Nair et al., 2023

Appendix 4. Interviews scripts

Interview 1: R.D-Head of BIM and Digital Assets

[...]

Interviewer

Yeah, sorry. Sorry for interrupting you.

Interviewee

That's right. And so, the BMS

Interviewer

It's Schneider.

Interviewee

Schneider's certainly one of them, and it's probably the main one that we've got. Yeah, the the.

Rich Draper (BIM and Digital Assets) 0 minutes 35 seconds

I don't think that it's doing anything specifically based on occupants' day.

Interviewer

Yeah.

Interviewee

0 minutes 43 seconds

But it is doing things based on.

Planned activity. Yeah. So, so.

Interviewer

OK.

Interviewee

Say a lecture theatre.

We'll have a planned activity in it.

And the timetabling team would pass information to the BMS team and the BMS team would make sure that the heating in that room is set right before the lecture starts. Yeah. And I'm sure there's an automated part to that. So, essentially the timetable schedule gets passed to the BMS team and the BMS.

Basically, plans turning the heating on for those scheduled sessions. Now that's problematic in some cases when the scheduled session happened. Doesn't happen essentially or nobody turns up and never has been heated for no reason. So, it does. It obviously has its flaws, yeah.

Interviewer

Yeah. Yeah, I was going to say.

Yeah. Does it consider historical dates or is it just based on how many people are expected to be there?

Interviewee

No, I don't think so.

Yeah, I believe so, yeah.

Interviewer

As a student, as a student, as someone who had to go and should big lecture theatres, especially for tutorial sessions, and now I will turn up that that would be great. A big issue within it.

Interviewee

Yeah, yeah. Oh, you know what? To heat to heat a massive room to get it to the right temperature. And then the lecture doesn't happen. Or, like, 10 people turn up.

To a 200-seater lecture theatre, it's a huge amount of potential heating loss.

Interviewer

So, in terms of value that the framework could bring.

That that would be quiet.

Interviewee

Oh, huge. Yeah. Because if you're if because you've got to consider right things so.

Interviewer

Yes.

Interviewee

Machine learning without the timetable isn't going to tell when somebody's going to turn up.

And the other thing is if let's say there's a lecture theatre and Tempe, let's say, let's say 100 people turn up to a lecture theatre, yeah.

Interviewer

Yes.

Interviewee

Only once it's triggered that there's 100 people there.

For it to turn the heating on that's too late. There's a heat. The room's only going to get to temperature by the time the lectures in it finished.

So, you still do need to have the planned bit, so you still need to have the schedule. We're expecting this to go ahead. Yeah. So, let's heat the room up. Ready for the lecture that still has to happen.

Now the machine learning aspect of it could be looking at historically, nobody's ever turned up to that lecture, so maybe we don't heat it up or better still, we don't heat it up and we reprogram the timetable to send those people to a small room. Yeah.

Interviewer

Yeah, exactly. Yeah.

Interviewee

Now, now you and I working in in the areas that we work in, if we got a ping on our phone. 20 minutes before the lecture to say don't go to that big lecture room, go to this one instead. I think we'd be OK with that. But there will be lots of other students across the university who won't be happy with that.

Interviewee

They won't be happy to,

Interviewer

It won't be feasible to certain extent.

Interviewee

No, because it there will be people cause, but people are people. We're not machines, unfortunately. So unfortunately, I guess depending on which way you look at it. You've got to remember, like in all of these different things. So, if you did get the text to, say, go to this room instead of this room and you're a person who's got, you know, significant anxiety or.

The issues or even? I mean, disabilities can be disabilities, is easy, not easy. But disabilities are OK because you can make sure that you're programming it into a room. If you've got a registered disability that you make sure you go to the right place. Because what we're working on at the minute is making sure our spaces have, like, a disability classification, so that there we know which ones are accessible. We know which ones are, like, close enough to toilets and stuff, and that's all. It's all things that must be considered for the individual student as well.

but that's OK, because that can be planned with machine learning, so you know big lectures to read it. But you've still got to think not everybody's just going to want to do that just because the class size has dropped below.

Drop below 60 people because we can fit 60 people into a significantly smaller room that's significantly cheaper to heat.

Significantly cheaper to light.

And frankly, you've got your lecturer and.

Who just will not pick up the ping 20 minutes before the lecture and they'll still go to the wrong place, and you will end up with chaos.

So, the earlier you can plan these things, the earlier you can plan using machine learning the better for the people of course. Drop below 60 people because we can fit 60 people into a significantly smaller room that's significantly cheaper to heat.

Significantly cheaper to light. And frankly, you've got your lecturer and.

Who just will not pick up the ping 20 minutes before the lecture and they'll still go to the wrong place, and you will end up with chaos.

So, the earlier you can plan these things, the earlier you can plan using machine learning the better for the people of course.

Interviewer

Yeah, that's something.

Interviewee

they're all of these things have got to be considered as well in in terms of the way it happens, because lecture still will happen and the lecture in the big lecture theatre. Will still happen even if there's only 10 people turn up. But those ten people are going to be freezing cold.

Interviewer

Yes.

Interviewee

Because we decided not to turn the heating on because there's not enough of them to to do it or we will do it and the university is going to lose lots of money because of the amount of heating that we do over the year. You know that kind of thing. there's all there's so many different aspects of this to consider.

Interviewer

In terms of value, obviously we'll talk about accuracy from one to five. Obviously, five would be the ability to shut down a a whole building.

Interviewee

Yeah.

Interviewer

As we as we discussed before.
What would 1 be in terms of value?

Interviewee

Close to the ability to see that you could do something, but not necessarily implementing it, so it's kind of presenting you from a value point of view. It's presenting you with the data to say you could have done this.

And it could have helped the university save some money. Carbon, whatever. I think from a value point of view.

I mean, there's it. I think if you look at, I mean zeros and I think isn't it.

So, zeros like we've got we've got no data, we don't care.

Interviewer

Yeah.

Interviewee

Will blindly carry on doing what we're doing.

Interviewer

So yeah, obviously if I come two days later and told you, yeah, you could have done this, but I wasn't able to give you to tell you to do that. So, in terms of time frames, the ability to effectively.

They did you do something and actually do something about it and five would be provide you with the data, all the information and act on it.

Interviewee

Yeah. Yeah, it's interesting that you said two days there. Is there a reason why you said two days?

Interviewer

Right. I'll tell you what two days is. the Siemens data, you know the heat maps that we talked about.

Interviewer

Yeah.

Interviewee

The data for those only appears two days later.

Alright, when you're when you're talking about?

Interviewee

The accuracy and the value. It's really good. The way that you've thought about that because one of the argument arguments or discussions, it is an argument. One of the discussions that we've had with semen is around how valuable their platform is.

I think it still does have value, but the issue is it doesn't have immediate value.

Because it isn't available straight away, it comes two days later, so it probably does fit on the scale of 1.

And then you look at the accuracy bit and it doesn't do the number of people; it just tells you that there's some activity happened.

Over a period, so it's probably on the scale of two.

Not necessarily one because it does tell you the time as well.

Interviewer

Yeah.

Interviewee

The time periods, it comes two days late, so if you're looking at and then I think your third one's cost isn't it and how much do things cost to implement?

Interviewer

I think that's going to be incorporated into efforts.

Interviewee

Effort. Yeah, yeah. Yeah. Amazing. Because

Interviewee

Like if we if we if we talk about the savings and so on. So, you've got.

I might get into a bit of trouble, but the Siemens, the Siemens initiative. Is you've got.

You've got you haven't quite got the accuracy.

Interviewer

Yes.

Interviewee

You haven't quite got the value bit in terms of. It's not immediate, it's not allowing you to do things straight away, so it's sitting on more or two, maybe three, you know, on your scale and then your effort.

Interviewer

I said it's 5, isn't it? It doesn't really.

Interviewee

Any effort is it's like massive effort, massive cost.

Interviewee

Yeah. So, we kind of if we use that your framework that you're putting together, if we were to Use that as a mapping exercise for any initiative.

That we go forwards with you end up with.

Maybe some questions as to why we did Siemens in the first place.

Interviewer

But if you don't, I if you don't mind, I'm asking. Why is it so? Why is it so, so costly? What is it? The sensors.

Interviewee

Why is it so costly? It's millions of pounds it well because we we're also changing the lighting as well. So, we're changing the lights from the standard lights to LED lights and they're putting the sensors in as well. So, the cost of the project is is has been significant because it's quite an impactful project. It's not, it's quite an intrusive project.

Interviewer

Is it intrusive in terms of? Does it bother the students?

Interviewee

Well, it has done, yeah. It it has done because we've we've had to take areas out of action to to do the light refit because we were essentially refitting the lights and the big one on the lights is to obviously change the old lights to new lights because swapping to LED lights is significantly better from a utility costing point of view. But the cost of the project, I don't know what the payback period on the project is.

Interviewer

Yes.

Interviewee

But the effort has been great.

So there has been a lot of effort, not just the cost as well, but the actual effort to do it because it's had to take areas of the university out of action whilst it's been fitted. We've got

constant meetings about you know the data and stuff. So, we are now getting insights there are I think one of the things that Siemens has done is it's been able to highlight. That there have been, I think they they've said they've got a list of like 37 rooms in those 26 buildings which haven't been used for six months. They've been able to highlight that to us. When we talk about the value. We may not be talking about immediate value all the time. the value from that point of view could be pushing up more to the three to the four into the value because it's now telling us spaces which we didn't know weren't being used and haven't been used for a significant amount of time. Six months is a long time for space. It's not to have been used.

Interviewer

Yes.

Interviewee

And we've now got a list of those rooms so we can then go and do something about, we can go and get them repurposed, turn them into something else, give them to people that are going to use them instead of the people that are currently occupying them.

Interviewer

So the value ranges from one because. It doesn't have any immediate value, but to three.

Interviewee

Well.

You might be wanting to look at two different value scales. One is short term, one is long term, so the short-term value.

Interviewee

Is the ability to react and affect things immediately in your long-term values. Your ability to then plan much better into the future.

To negate the value conversation. So, it might be worth having a look at that. Whether you can do that in a single scale or whether you need that as two separate scales, I don't know. Because you can't look at Siemens in isolation and say, well the value is 1 because it doesn't tell us the answer today.

[...]

[CONFIDENTIAL INFORMATION]

[...]

Interviewee

And we haven't since re-advertised or re-established how much money we're going to spend on a capital programme because we just haven't worked it out because of the way that the money's kind of not working in the university so.

Nowhere else going with that. Oh yeah, it's because I was talking about it because we can't get. Basically, we can't just say like Trevor, my director Trevor. I need another man to come and do some work on tandem. He's not just going to say yes, whereas five years ago he might have said yes. Absolutely. This is a fantastic thing.

Interviewee

It's save us loads of money. Just have a person doing it, but we're not in an environment where we can just flippantly kind of say let's just bring in a new person because it it's not working like that. And that's why we've not been able to progress it the same way and that's why it's been a bit of a hobby project for us. But we are now more and more using it in our day-to-day work. So, the images that I showed you just now, all of that's come from tandem because I can use tandem to manipulate data in a way that I can really then just present it to people nicely. So, we're getting more and more use out of it and the more we use it.

Interviewer

Yes

Interviewee

What's it doing? More and more stuff with it, so we'll start being able to bring in sensor data and we'll start be able to start to use this as a communication tool.

To change the way the universities operating and change you know, really affect it and use that machine, learning that you're talking about and actually build this in and actually get some real change happening. And this is really going to be a platform that's going to allow us to do that so.

Interviewer

The budget's going to. It's definitely going to be a challenge for the implementation of that of that that framework, wouldn't it? Apart from obviously.

Interviewee

Yeah.

Interviewer

I imagine that you don't have the trained personnel for it as well. do you have people that can design these algorithms and?

Interviewee

No, no, certainly not within my team, Tim, Tim's probably closer to it, but.

Interviewee

He also he he just doesn't have time to do this. He hasn't got the resource to do it. Hopefully he's building up a bit more of a team to be able to help do that so.

Interviewer

Yes.

Interviewee

Long term, but the vision, the vision from the university's point of view and from an estate's point of view Is smart campus is absolutely the direction we've got to go in.

Rich Draper (BIM and Digital Assets) 23 minutes 24 seconds

And Trevor, our director, is absolutely, I don't know if you're following on LinkedIn at all at the minute. Probably worth it.

Interviewee

Trevor,

Interviewer

Trevor Payne?

Interviewee

Yeah. Trevor Payne's our director. Yeah. If you follow him on LinkedIn, everything he's talking about is smart campus type stuff and prop you know, prop tech and new technologies and stuff like that.

Interviewer

Yes, I follow him.

Interviewee

Really up for doing all this kind of stuff, but again constrained by budget. And there's only so many ideas we can have in our head at once.

Interviewer

Yes, definitely.

True, that's true, especially if that's just a hobby for you. I mean, I thought that was something that, you know, the actual university was working on.

Interviewee

Yeah. it is by virtue of the fact that we're doing it.

Interviewer

But it came from your efforts. So if you didn't go that way, they wouldn't have followed you.

Interviewee

If we didn't do it, we wouldn't be doing it, no, for sure.

Interviewer

It genuinely was, I think myself and Matt and Tim, we were sat in a room. We went to go, and we did a visit to Autodesk they basically showed us a product and over the lunch break we got it working and started putting some of our models into it. So yeah. Sorry.

Interviewer

Just so that I don't want to take too much of your time just for the metrics, can you just give me just some values for efforts value and accuracy?
For the Wi-Fi.

Interviewee

For Effort 5 is going to be a top value effort. It's going to be, it's going to be minimal effort, which is going to be, let's say a couple of weeks' worth of work, right to get to get something working, and obviously essentially what is staff cost, just the cost of my time whereas the all the way down to year one is going to be.

£1,000,000 projects.

That's going to take a couple of years to deliver.

It's going to be somewhere between those two, yeah.

Interviewer

No, no, I meant for the Wi-Fi as a metric.

As a metric to monitor occupancy, how would you rate the efforts involved is that would that, would that be 2?

Interviewer

And what and for value would you say it's around 3 or 4?

Interviewee

Yeah, I'd say.

Interviewer

How much value does it give you?

Interviewee

I'd say, well, it did give us a load of value, but I'd say 4.

Because if we we're going to refine it more, we've we well we when we found some time we'll refine it more so that we can go down to space level rather than just building level.

Interviewer

Yes

Interviewee

But for what it was doing, It was relatively high value, yeah.

Interviewer

And accuracy would that be, I'd say two or three, right? Because it doesn't really give you the amount.

Interviewee

I'd say three. It gives us. It does give us a number.

But it doesn't tell us exactly where they are in the building, and that's the issue that we've got there.

Interviewer

for my attendance.

Effort 1 from your end.

Interviewee

Zero from our point of view because.

Yeah, do one because the university have effort to do it

And then value, I would say probably three or four, and the reason is because it isn't wholly accurate. There's a thing called WhatsApp groups and when the QR code is provided by the lecturer, that often gets provided onto WhatsApp groups and everybody who isn't there also registers.

Interviewer

I will not confirm that.

Interviewee

No, no.

Interviewer

Accuracy. Well, based on that, I would say accuracy would be two to three depending on how students...

Interviewee

Yeah, I would say so, yeah.

Interviewer

For or the framework I'm proposing, It's going to take quite some effort in terms of the cultural change the course involves and the time that you would need to dedicate to it. So, I'm going to say that the effort's definitely going to be quite, quite high, so maybe 4.

Interviewee

Yes, I agree.

Interviewer

5 in terms of value given its potential and obviously accuracy as well, is going to be 5 but it will depend on how well the model is designed.

You can play with it. If you just need a level of accuracy of one or two or three or four, then you can choose what level you need depending on the algorithm. So that's one of the good things about this framework. The accuracy is depending on your needs basically.

Interviewee

Yeah, absolutely.

KI

Interviewer

And yeah, would you say that's something that the University and the estates team would be interested in? And do you think that's worth researching more on in the future?

Interviewee

Yes, yes yea, I think so and I think there probably will be companies like Autodesk that would be interested as well. So would worth you know if you can get some stuff going through tandem like that. I think Autodesk could be really interested in it.

Interviewee

Perfect. That's all. Thank you for your time.

Interviewee

Alright, good.

Interviewer

thank you so much for your help. Really appreciate it.

Interviewee

If we need another session, let's just do another session as some point as well if you need to.

Interview 2: M.H-Data Manager

Interviewer: Can you first just give me your role on the estates team, if you're comfortable with that, and how many years of experience you have?

Interviewee: Sure. My job title here is Senior Digital Information Manager. I work within the BIM digital asset team. Obviously, you've spoken to R.D. and C.P. who are also on the team. We look at anything BIM on projects, how we can use that information operationally, space management, and support the data aside. So, I've been at the University for three years this June, in a similar role. And before that, I was at the University of Worcester, focusing more operationally, looking at the CAFM systems, all the help desk or asset data, and looking after floor plans. I was there for five years. I've been in the Estates industry in the university, coming up to eight years now. And I also did a data analyst level four apprenticeship. So that really got me into not just looking at the data on systems, but how do you analyse that data, which I think comes into a lot of what we'll talk about going forward.

Interviewer: We've got a good foundation because we've got our sort of been requirements for any sort of project and capital project, major referbs, medium size referbs as well, or sort of the new buildings. We always get this BIM model, which, because of the structure we've always had, we've got goods of asset data in there. We know, in a building base of the data, what's a boiler and what's a wall, what's a door, for example. We've got a good foundation, and we've been looking at in the past how we then use that, along with the other systems where we got information. Every building's got the BMS system but in the past, they've been two very separate systems. Even in projects, we'll speak to one group of people who sort out data, but the same elements will be in the BMS. They will have completely different references. There's no correlation between the two. So, over the past few years, we've been looking at, how do we then start bringing all this data together? Because that's where you get the real insight, not have to look at a building and go to BMS. How do we bring it together? So, we had to look at a few digital twins, but about this on last year we got our hands on tandem, that allowed us to put our BIM models in, start viewing a, viewing the data in there, sort of classification. And then that also allows us to stream in the information. What we did initially was working with our IT team, begin to stream in WIFI data. So, the number of WIFI devices roughly indicates the number of people in the building, and then use and off see the areas as well work. What meter squared each person in that building has? Because when we put the people in, for example, you've got the library that was showing a thousand people, because it's one of the busiest buildings on campus, whereas this building here, let's say West, was probably showing 15. So, on a heat map, one's looking green, one's looking red, but in reality they're both being used, all right, because they're different sizes. That's why we took it down to meter squared. So that's all good because we can start thinking space-wise, right? We could shut down this building because no one's using it. So, we did that for a few buildings, but what our next step is now is to see how we can get that BMS data into tandem, to then view that heat map across our campus. Which Tandem have now bought, our Tandem Connect. Again, it's probably down to, down to funding. And the way our team is in estates, we don't really have the coding experience, which is, so if you need to sort of how to create these data integrations. But we're hoping that this time and Connect allows us to have solve an easier way into that, of getting the data connections to then get a real value, bring all the data together.

Interviewer: In terms of metrics, what are you using? You've mentioned WIFI. C.P. also mentioned you using the attendance from MuUoB app and in collaboration with Siemens, lighting sensors. Can you tell me more about these three metrics? How effective are they? Are you comfortable with them or are there other types of data streams that you're looking at?

Interviewee: We did a task around October last year. And we went and did some manual counts of some teaching rooms across the week. And we compared that to the timetable, so how many people were meant to be in those rooms? Then we compared it to my attendance. And what we are still waiting to do, just because we undergo the WIFI data, is then relate that to the WIFI, to then get sort of a good level of accuracy across here. And we also did the Enlighted data, sir. Obviously the same as data. So, if I go through each one, though, so my attendance, when we did the comparison to manual counts, it overall worked out very well. Or there were some quite big differences in the numbers. And what we think that was due to potentially, obviously someone going to a lecture, sending the QR code out, then on WhatsApp, whatever, to other people, and then signing into a lecture without being there. So, it was good overall, but there were some definite differences, so we couldn't be homes and contents in that. Wi-Fi, we get to really do checks just due to access the data. And the Siemens one is it's good. So, in a room where there is no one, that light and it's on a hundred percent. But as soon as someone goes in, we've got motion detectors on it, and it brings it down to about twenty percent. So, from the Siemens data point of view, where we got the sensors, it's quite a good indication, but there are some rooms where they'll put it on, but they're not really using it. Or the other way around, is they're using it, but because the room's so small, and the sensor might be a bit far away from the occupancy area, it's still putting it on and off. So that's a bit of a challenge to them. But I think as I alluded to earlier, once we get the bin data in, which is quite complex because of different formats, but we're hoping when you get to that level, you get a real good indicator of where people are on, not even at a person level, you can start looking at what we're doing now is looking at the cafeteria spaces and see how many people are in that space, and then using queuing theory to understand the sort of capacities and people moving in and out of spaces. So, we can start putting different scenarios into the room, which is quite good, but it is still early days.

Interviewer: So how do you evaluate the value, effort, and accuracy of these metrics? And then do you have any recommendations of how to further develop the metrics or indeed, to further help you get a better understanding of occupancy in the space and indeed, energy usage in a space.

Interviewee: Yeah. The attendance data has been relatively good. It's been relatively easy because there's no outlay to get that data. It's quite a simple format. but we can't go through and manually count and understand the data. We had a pilot run at doing this, I think it was November last year. And we started looking at, so we've obviously got lectures, but we've also got tutorials. And we've got other different types of sessions, but the way the systems were set up, all the locations were set up, it was difficult to pull them out of the attendance. So, we did a pilot. We've got a good group. I think it's probably seven or eight people from different parts of the university. We sat down and we started looking at what sessions we got. And then we had to break it down into what would we want to monitor. And then, like you said, so you got to break it down into some key things, but the accuracy of the metrics. it's quite a difficult one, really, because we've got attendance, which is good. But then you got to cross-reference that to where they should be in that time. But then that's not a true accurate measure of what's going on in a space. Because what we're interested in is, is there a person there or not, or is

someone occupying that space? But I think, as I mentioned, hopefully when we get the BIM data in and it's very complex because of all the different formats, that will allow us to really understand it. I do think it's quite difficult. You got to go through the steps and sort of break it down into what you can manage. And then start looking at how you can then improve it going forward. But I do think it's quite difficult. And it's probably not something you're ever going to get a hundred percent. And that's why we're looking at different ways to monitor the space. And that's sort of led us into the queuing theory.

Interviewer: can you identify potential challenges in implementing new frameworks such as incorporating AI algorithms for real-time data analysis?

Interviewee: So, one of the main challenges we've got, which I'm sure you've heard of, is funding, but, as I mentioned, we've got a good foundation with the, with the bin data in there. But what we've found over the last few years is we, we get these data streams, but we don't necessarily have the resource to look at them and understand them. A few years ago, we did a trial with our BIM systems and camera analytics, which we could stream into this platform. I think it was probably two weeks where we had a trial of this software. So, obviously, my role is digital information. I was the sort of digital champion for the software. And I had to go out and speak to all different people, faculties, other services. And I didn't really know what I was looking for. But there's obviously some key things I was looking at, but there were also some other things. So, we had security who wanted to know if someone was lying on the floor, then to be alerted. obviously, that's a key thing, but I never would have thought of that because, from my point of view, I'm not thinking security. I think one of the biggest challenges we've got is resource allocation. We do have a pilot coming up in the next few months where we're looking at another digital twin platform, which we can start looking at some of the people moving through spaces, and we're going to put some sensors on them and understand how they're moving.

But again, it's very resource heavy. we've got to pick our battles on that. And then another challenge we've got is probably more cultural change. I think people are obviously used to doing things a certain way. And I think when you introduce new technologies, people are going to resist change. it's trying to work with people to understand how we can take this forward. But I think if you've got good evidence, which we have from the last few years of projects we've done, you can then start to bring people with you. But I think it's a challenge. And I think it's probably an ongoing challenge as well. So yeah, I think there's a few challenges there. But if you can get some good evidence behind you and then just try and start on a small scale and then build it up, I think you can start to tackle some of these challenges. it's not easy. And I think you've got to have a good team around you as well and be willing to learn and understand and be willing to change as well. there are a few challenges there. But yeah, it will lead to good outcome, it has a lot of potential.

Interviewer: Thank you for your time.

Interview 3: C.P-Space Manager

Interviewer

Hi, my name is Khalil.

Thank you for being here and agreeing to answer a few questions related to my research project. Before starting can you just tell me what your role is on the Estates team?

Interviewee

I am responsible for space management.
All things related to space.

Interviewer

OK. And how many years of experience do you have roughly?

Interviewee

I've been in higher education for 26 years.

Interviewer

Yes.

Interviewer

On how to implement artificial intelligence in BIM-FM. After talking with R.D. is to look at how AI technology can be used for occupancy monitoring, detection, and prediction for energy optimisation. From what R.D. told me is that you guys use Wi-Fi to monitor occupancy and from there, you look at how to manage a space more efficiently which helps you make decisions on how to optimise energy consumptions, right?

Interviewee

That's one of the data streams that we're looking at, yes.

Space in the traditional sense is owned and managed by the university, so we will work with the colleges and the professional services to understand how the space is being used and if a new research bid comes in or a new project comes in, then we will look at the spatial requirements for it. So, we hold all our space data currently on our Revit files and the space utilisation piece is a fairly new piece, although it's something that within the HE sector has been scrutinised and analysed for years and years with minimum effectiveness, should we say so we report key performance indicators for a number of different areas, one is space utilisation, so we report into that adhesive statistics, which is the estate management reporting.

And we would have historically reported on teaching space. So that is one of the metrics. Which across the universities we've realised that the methodology is fundamentally flawed because it's open to interpretation. Birmingham, I think from other different universities are probably ahead of the game because of some of the initiatives that we're working and so one of them is that the Siemens initiative. We're looking at the Enlighted data, so we've started getting.

Information through on our 25 different buildings where we've got the Enlighted sensors installed. We're still trying to set the parameters for that and we're still testing that. So, the reason that the Wi-Fi data had been fast tracked is because we want to validate and triangulate the Siemens data as well. So, we're pulling in various data streams, which will be Wi-Fi. It

could be my attendance. So that is in a teaching space where students will check in. So that is a fairly accurate figure depending on when the check in window is set for which across the different colleges there's a different time frame. If it's within sort of an hour or half an hour of the teaching session, it's accurate. If it the check in Windows out for 24 hours, it may not be so accurate because students afterwards couldn't manually check in.

But then we're pulling these three data sets through together to analyse well, how efficiently are ours the basis being used? So, it's still fairly early days and what we're doing.

Interviewee

But it has been helping with some of the management decisions. So an example with the Siemens data that there are certain spaces that although we can't understand what the true utilisation figure is, we know if rooms are being used so that the frequency figure is enough for us to start having those conversations to say well actually the space is not being used, it's allocated to an academic office, but it hasn't been used for six months.

And there has been the odd case where actually that information has led to a different management decision.

So that's the three key metrics that we're looking. An occupancy figure and multiply 1 by the other to get the utilisation figure and that is that the heater definitions that sector wide we all use and abide by or should at least.

Interviewer

So, you said frequency times what? Sorry.

Interviewee

the frequency multiplied by the occupancy.

Interviewer

OK so.

Interviewee

So how the frequency is how often a room is used. The occupancy is when that room is used, how full it is and that then gives the utilisation figure.

Interviewer

Using the three the data streams that you you've identified Wi-Fi and my attendance.

Is that is that accurate enough for you to make decisions, or does it have any limitations or challenges that you face?

Interviewee

Well, I mentioned my attendance depending on when that check in.

Interviewer

Yeah.

Interviewee

Window is and that I think is something that the university will decide and define as part of the policy of it, so that. It's probably as accurate as it can be, but we'll see and it's enough for us to be able to start scrutinising certain spaces.

Interviewer

There's a there's a very important point because on a meeting with R.D., we have identified 4 key factors to evaluate the air techniques that I'm that I'm working on for occupancy monitoring. So, we have identified value efforts, cost, and accuracy and what I want to focus on here is accuracy. How would you define accuracy and what is not for accuracy because obviously there are techniques that have. A maximum level of accuracy to detect and predicts occupancy, but you don't really need that. You do. You only need to know. An average of the occupancy that is in the building to decide how much is enough to decide.

Interviewee

I think it depends on the space type, so whether you're looking at.
A research space, A teaching space, or a supporting type of space.

Interviewer

So, if you were about if you were to talk about a building that the occupants of a building. What time frames would you work on a daily time frame or hourly or weekly? How much? How is time important here? How?

Interviewee

Again, I think it depends on the space type being able to get occupancy data from between 9:00 and five I think is important.

Interviewer

9 to 5. Would daily input of data be enough, or would you need hourly or? Or weekly.

Interviewee

I think for teaching space we need hourly.

Interviewer

Yes, OK.

Interviewee

So, it depends on the activity that takes place in the space. So, if you consider a research area, it might be a research lab that is has got equipment in its running 24/7.
And it might show that it's not occupied, but it perhaps doesn't need to be occupied.

Interviewer

So, if I were to propose a framework, For an AI driven bim-fm system. With AI techniques being used for occupancy detection and prediction. As a space manager, what requirements would that new framework need or require? In terms of accuracy and value Effort for implementation and its cost as well for implementation.

Interviewee

Can you give me a little bit more context?

Interviewer

If it requires too much effort and low value, that would not be great for you, right?
So, what would be the optimum level of?
Value and efforts.

For the implementation of that framework and how much would, how important is cost for you?

Interviewee:

I suppose it depends on how you define value and how you define effort. How you define cost.

Interviewer

The value is. I define it as. How much benefit? How much this framework is helpful to you.

Interviewee

OK.

Interviewer:

In terms of energy optimization and space management and the effort is how much effort is required from you to implement this this new framework.

Interviewee

So when you use the term framework, actually what you're talking about is a system, not a framework. Is that correct?

Interviewer:

Yes, a system.

Interviewee

It depends on the value of the product at the end of it.

It's very hard to say because what you're talking about is hypothetical.

Interviewer:

Yes, that's true.

It's only hypothetical.

Interviewee

What you'd need to consider is.

How would all of the systems feed into each other and the alignment of those systems? And the challenge there would be the cost of it all.

Interviewer:

I've been working on a digital twin of the old gym using Autodesk Tandem. That's what you use, right? Now.

Interviewee

Yes

Interviewer:

I've been on a meeting.

Last week, on Tandem Connect.

Which will be used to connect all these systems together through a digital twin.

Do you have a plan to use tandem connect?

Interviewee

Now, we're pulling the data streams into power BI. That's how we're looking at the data now.

Interviewer:

And how effective is this?

Interviewee

It's effective because it's easy to achieve.

Interviewer:

Have you? I've identified any limitations or any challenges when using the system?

Interviewee

Early days but. Not really, not at the moment.

Interviewer:

OK.

Yeah, that should be all.

Is there anything else that you would like to add in terms of Space management.

Or do you have any suggestions for an AI driven BMFM framework to be hypothetically proposed.

Interviewee

I'm interested in looking at your title.

You say a novel framework.

Interviewer:

Yes.

Interviewee

Define that.

Interviewer:

So, at the university, they haven't been any AI technology being used to detect.

Or predicts occupancy or just to be integrated in any BIM FM system. At the beginning, my initial idea was to look at how AI can be implemented in BIM in general, but obviously after research, I've realised that there's a lot of potential in in facility management. So why call it novel is because currently university there hasn't been any.

Attention giving to AI in terms of occupancy monitoring or facility management.

Interviewee

OK.

And have you had wider conversations with other universities?

Interviewer:

No, not really, no. I've I've looked at.

Interviewee

Is that is that your intention as part of your research?

Interviewer:

As a PhD, maybe as a masters know.

Interviewee

Ah, OK, so you're thinking of taking it to the next level?

Interviewer:

Yes

I would like to, yeah.

Interviewee

OK.

Interviewer:

Exploring what kind of AI techniques can be used and what kind of future the research direction there will be. Obviously, if we're talking about. A small campus right now using AI. Well, what would that mean for smart countries or smart cities in the future? That's what I'm in the future, a possible future research direction I'm looking at.

Interviewee

That sounds good. OK.

Interviewer:

One last question, if that's OK.

Whether detection or prediction I was thinking about something the other day on Wednesdays, for example, its sports day, right? And there aren't many lectures even Friday afternoon. There aren't many lectures. Are this the kind of stuff that you look at? So historical data?

Interviewee

So, for the teaching spaces, we would exclude the Wednesday afternoon as part of our data analysis for that very reason, we wouldn't exclude a Friday afternoon because in theory we should be utilising our teaching spaces during that period.

But yeah, I'd say this, I think. You need to break it down by space-time because space is used so differently across the campus. Teaching spaces we've got, we understand what the, the norms and the parameters are research spaces not so easily.

Interviewer:

And why is that? Why is research?

Interviewee

Just because of the nature of what happens we can capture the space data so we would know if it was a research lab with fitted benches or there was a wet lab or an E lab, but we wouldn't understand the activity that takes place in it.

Necessarily, without doing further analysis, so those spaces have probably got equipment running in there, but they don't need people in there to be able to understand utilisation because the space is supporting the research equipment.

Interviewee

The one of the techniques I'm exploring is computer vision, so using cameras.

Interviewee

OK.

Interviewer

Would that be ideal regardless of the privacy issues for research areas because you will be able to see and understand. What is going on exactly? How is space is being used?

Interviewee

A lot of universities for utilisation of teaching spaces have cameras in. They have cameras in as part of the support for lecture capture.

And you can tap into those feeds to understand how those space is being used. So, from the teaching perspective it's easy. It's scalable because there is a genuine requirement for it, but not so much in research spaces and not for academic office as well because they're clearly privacy issues there.

Interviewer:

Yes, absolutely.

Just one last thing. Apart from historical data and Wi-Fi.

And attendance? What other metrics do you use. Have you looked at any in any kind of environmental data, CO2 for example. Or Air quality.

Do you have any sensors being used for these types of data capture?

Interviewee

Yes, we do as part of the wider Siemens project.

And I know that other universities have used CO2 monitoring to understand utilisation.

Interviewer:

21 minutes 29 seconds

Is that something that you use?

Interviewee

And then you've also got card access data as well.

So, if you've got a controlled environment, you can work out how many people are coming in and going out, but a lot of the space is across campus. Here you've got multiple entrances and not all of them have got code access on them. And again, you've got the under the data protection requirements that sit around that. So, we have to be very careful.

Interviewer:

Yes

Interviewer:

Thank you so much for your help and for your insights. Just one last time. Is it OK for me to use all the information you've given me in my reports?

Interviewee

It is. I will get the documents signed and sent back to you.

Interviewer:

Perfect.

Interviewee

A little later on today, if that's all right.

Interviewer:

Yes, of course, no worries at all.

Interviewer:

I'm not sure how the recording here works. Will I automatically get a copy of the?

Interviewer:

Of this meeting, or how does it work?

Interviewee

You should do when the meeting has ended, you will be able to see the transcript.

Interviewer:

Yeah. Would you be interested in getting a copy of my research paper when it's all done and submitted?

Interviewee

Of course I would, yeah.
What's your time frame?
Oh OK, imminently then.

Interviewer:

Yes. One month.

Interviewee

Well, good luck with it.

Interviewer:

Thank you very much. You too good luck with your research.

Interviewee

Well, I'm now on year six, so it's a very slow progress. I'm nearly there.

Interviewer:

Well, good luck.

Interviewee

Alright, thank you.

Interviewer:

Thank you again for your help. Bye.

Appendix 6: Reflection

My final year project has been a challenging journey, marked by many challenges and obstacles along the way. Personal circumstances affecting my mental health limited the amount of time I could dedicate to my project, which was unfortunately reflected in my interim report. However, the constructive feedback that I received allowed me to work produce a better piece of work this semester, which reflects better my abilities. With determination and perseverance, and despite the challenging times I went through, I have done my very best make significant progress.

The feedback highlighted several issues with my interim report:

- Lack of clarity in the scope of study which was too broad.
- Too broad discussion of key aspects.
- Unreliable and inadequate references.
- Absence of state-of-the-art considerations.
- Format was not always respected.

Therefore, I have defined a more specific scope for my study. I have designed a theoretical framework focusing on machine learning applications for occupancy monitoring in BIM-FM. In my literature review, I have dedicated more time in exploring the various aspects of my research such as BIM, BIM-FM, and AI. I used a systematic literature review to identify state-of-the-art applications of AI for occupancy monitoring and reported the results in a consistent way. I have also paid more attention to format requirements, ensuring correct referencing by using Mendeley.

Throughout this process the support and assistance of the Estates team, especially from R.D. C.P. and M.H., were invaluable to my research efforts. Their experience and guidance allowed me to gain invaluable knowledge which motivates me to take my research further.

I am also deeply grateful to the unwavering support and guidance provided by my supervisor, Dr Richard Davies, who not only offered academic mentorship throughout the year, but also provided invaluable personal support, which I greatly appreciate.

In conclusion, given my circumstances, I have done my best to produce a piece of paper that reflects my capabilities. I have gained invaluable insight, and it has given me many opportunities such as getting involved with the Estates team at UoB. This research also motivates me to do another master's where I will be working more in depth in designing the AI-driven occupancy monitoring model which I am hoping will help the Estates team in their work.